Polynomial Secret Sharing Schemes and Algebraic Matroids

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Abstract. Polynomial secret sharing schemes generalize the linear ones by adding more expressivity and giving room for efficiency improvements. In these schemes, the secret is an element of a finite field, and the shares are obtained by evaluating polynomials on the secret and some random field elements, i.e., for every party there is a set of polynomials that computes the share of the party. Notably, for general access structures, the best known polynomial schemes have better share size than the best known linear ones. This work investigates the properties of the polynomials and the fields that allow these efficiency gains and aims to characterize the access structures that benefit from them.

We focus first on ideal schemes, which are optimal in the sense that the size of each share is the size of the secret. We prove that, under some restrictions, if the degrees of the sharing polynomials are not too big compared to the size of the field, then its access structure is a port of an algebraic matroid. This result establishes a new connection between ideal schemes and matroids, extending the known connection between ideal linear schemes and linearly representable matroids.

For general polynomial schemes, we extend these results and analyze their privacy and correctness. Additionally, we show that given a set of polynomials over a field of large characteristic, one can construct linear schemes that realize the access structure determined by these polynomials; as a consequence, polynomial secret sharing schemes over these fields are not stronger than linear schemes.

While there exist ports of algebraic matroids that do not admit ideal schemes, we demonstrate that they always admit schemes with statistical security and information ratio tending to one. This is achieved by considering schemes where the sharings are points in algebraic varieties.

This work is the full version of the paper presented at TCC 2025 [15].

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1 Introduction

Secret sharing schemes are a cryptographic primitive designed to protect a secret value by distributing it into shares. In these schemes, the secret is held by the *dealer*, and each share is sent privately to a different party. A subset of parties is *authorized* if their shares determine the secret value. The *access structure* of a secret sharing scheme is the family of authorized subsets. A scheme is *perfect* if subsets of parties that are not authorized cannot obtain any information about the secret, in the information-theoretic sense. Secret sharing schemes were introduced by Blakley [23] and Shamir [76] in 1979, and are used to prevent the disclosure or the loss of the secret value in many different cryptographic applications (see [10], for example, for a list of applications). For these applications, there is a need for efficient schemes and, in particular, shares should be as short as possible. This leads to the problem of minimizing the share size (or the *information ratio*, which is the ratio between the size of the secret and the size of the largest share).

The most common secret sharing schemes are linear, e.g., [53,20,29,57]. However, most access structures require linear schemes with exponential share size [6,13,73]. This motivates the study of other families of schemes that can overcome this limitation. To address this goal, Paskin-Cherniavsky and Radune [72] defined secret sharing schemes with polynomial sharing; in such schemes, the secret is an element of a finite field and the shares are obtained by applying polynomials to the secret and some random field elements. These schemes were further studied in [17,14]. Polynomial secret sharing schemes generalize linear secret sharing schemes, adding more expressivity and giving room for more efficient schemes. For general access structures, the best known polynomial schemes have better share size than the best known linear ones [17]. The motivation of this work is to identify the properties of the polynomials and the fields that allow this efficiency gain, and characterize the access structures for which it is possible.

1.1 Our Goals

In a linear secret sharing scheme, each party is associated with a set of vectors, i.e., the i-th party is associated with some vectors $v_{i,1} \dots, v_{i,k_i} \in \mathbb{F}^{m+1}$ for some finite field $\mathbb{F}$ and $m > 0$. To share a secret $s \in \mathbb{F}$, the dealer picks uniformly random elements $r_1, \dots, r_m \in \mathbb{F}$ and sends $v_{i,j} \cdot (s, r_1, \dots, r_m)$ to the i-th party for $1 \leq j \leq k_i$. Namely, each share contains k_i elements, and each element is a linear combination of the secret s and the random elements $r_1, \dots, r_m$.

In a secret sharing scheme with polynomial sharing, or *polynomial secret sharing scheme*, the secret is an element of a finite field $\mathbb{F}$, and the shares are obtained by applying polynomials to the secret and some random field elements. That is, the i-th party is associated with some polynomials $P_{i,1}, \dots, P_{i,k_i} \in \mathbb{F}[x_0, \dots, x_m]$ for some $m, k_i > 0$, and the shares of a secret $s \in \mathbb{F}$ are generated by picking uniformly distributed randomness $r_1, \dots, r_m \in \mathbb{F}$ and sending $P_{i,j}(s, r_1, \dots, r_m)$ to the i-th party for $1 \leq j \leq k_i$.

Our first objective is to understand when polynomial schemes provide an efficiency gain with respect to linear schemes. For that, we analyze the degrees of the sharing polynomials and the fields that allow this efficiency gain.

Then, we study the access structures that admit efficient polynomial schemes. Recall that linear schemes are perfect and that their access structure is determined by the vectors of the scheme: A subset of parties A is authorized if and only if the vector $e = (1, 0, \dots, 0)$ depends on A's vectors. Hence, the search for a linear scheme for a given access structure can be translated into a linear algebra problem. In contrast, we lack a systematic approach for constructing polynomial schemes for an access structure.

Our second objective is to provide a criterion for determining the access structure of perfect polynomial schemes. First, we study ideal schemes with polynomial sharing. In an *ideal* secret sharing scheme, the size of each share is equal to the size of the secret; this is the best possible situation for a perfect scheme [58]. In this case, we say that its access structure is called *ideal* as well. We aim to characterize the access structures that admit ideal polynomial schemes.

For the study of ideal schemes, we use results in matroid theory and algebra. A matroid is a combinatorial structure generalizing linear spaces and cycles in undirected graphs; a matroid is a pair (E, r) where E is a set of points and $r : 2^E \to \mathbb{N}$ is a rank function satisfying some axioms (see Definition 2.5). Brickell and Davenport [30] proved that ideal access structures are determined by matroids, i.e., the minimal authorized subsets of an ideal access structure correspond to the circuits of a matroid (i.e., a minimal set A such that $r(A) < |A|$) containing the point 0. In this case, we say that the access structure is a *port* of that matroid. In contrast, not all matroids determine ideal access structures [75]. The characterization of matroids that determine access structures with ideal schemes is an open problem. This connection between secret sharing schemes and matroids has been crucial in the search for positive and negative results on the existence of efficient secret sharing schemes, e.g., in lower bounds on the share size for general access structures [13,66], characterization of ideal multipartite and ideal weighted threshold access structures [18,45,46], and separation results for different secret sharing techniques [12,16,11]. For example, it is known that an access structure has an ideal *linear* secret sharing scheme if and only if the access structure is a port of a *representable* matroid [29]. We would like to extend this tight connection to the families of ideal polynomial schemes and matroids determined by polynomials.

Our third objective is to determine the security properties of schemes with polynomial sharing. Polynomial secret sharing schemes are not perfect, in general. Some subsets of parties may have partial information about the secret (i.e., they are neither authorized nor forbidden), as shown in the following example.

Example 1.1. Consider the scheme with three parties in a finite field $\mathbb{F}_p$ with $p > 2$, where the share of the first party is $P_1(s, r) = r$, the share of the second party is $P_2(s, r) = s^2 + r$, and the share of the third party is $P_3(s, r) = s + r^2$. It does not have perfect correctness because the first and second parties cannot fully recover the secret, i.e., they can determine s^2 but may have two options for the secret s. Moreover, the third party has partial information about the secret: that is, there are $\frac{p+1}{2}$ possible values of the secret. However, note that the same polynomials over the finite field $\mathbb{F}_4$ give a perfect secret sharing scheme. △

We analyze the properties of these schemes and provide methods to transform them into schemes with strong security guarantees. We show that the algebraic dependence of the polynomials is important in achieving the above goals. We say that a set of polynomials $\{P_1, \dots, P_k\}$ with $P_i \in \mathbb{F}[x_0, \dots, x_m]$ is algebraically dependent over $\mathbb{F}$ if there exists a nonzero polynomial $Q \in \mathbb{F}[y_1, \dots, y_k]$ satisfying $Q(P_1, \dots, P_k) = 0$. This polynomial Q is called the annihilator of $P_1, \dots, P_k$. We say that a polynomial P_0 depends on a set of polynomials $\{P_1, \dots, P_k\}$ if there exists an annihilator of $\{P_0, \dots, P_k\}$ that depends on the first coordinate.

Example 1.2. Consider the polynomials $P_0(x_1, x_2) = x_1$, $P_1(x_1, x_2) = x_1^2 + x_2$, $P_2(x_1, x_2) = x_1 + x_2^2$, and $P_3(x_1, x_2) = x_1 x_2$ over $\mathbb{F}_4[x_1, x_2]$. Observe that P_0, P_1, P_2 are algebraically dependent over $\mathbb{F}_4$ because $Q(y_1, y_2, y_3) = y_1^4 + y_1 + y_2^2 + y_3$ is the annihilator of P_0, P_1, P_2 since

$$Q(P_0, P_1, P_2) = x_1^4 + x_1 + (x_1^2 + x_2)^2 + x_1 + x_2^2 = 0.$$

The polynomials P_0, P_1, and P_3 are also algebraically dependent since the polynomial $R(y_1, y_2, y_3) = y_1^3 + y_1 y_2 + y_4$ is an annihilator for P_0, P_1, and P_3 as

$$R(P_0, P_1, P_3) = x_1^3 + x_1(x_1^2 + x_2) + x_1 x_2 = 0.$$

Consequently, P_0 is dependent on $\{P_1, P_2\}$ and on $\{P_1, P_3\}$. Analogously, it can be proved that P_0 is dependent on $\{P_2, P_3\}$. △

1.2 Our Results

Our work unveils algebraic properties of polynomial secret sharing schemes that are important for characterizing their access structure and understanding the efficiency gain provided by these schemes. In a scheme with polynomial sharing, we can define an access structure Γ by the algebraic dependence of the polynomials as follows: A subset is in Γ if the secret, i.e., the polynomial $P_0(s, r_1, \dots, r_m) = s$, algebraically depends on the polynomials of these parties.

Example 1.3. Consider the polynomials P_0, P_1, P_2, P_3 over $\mathbb{F}_4[x_1, x_2]$ from Example 1.2. The access structure defined by the algebraic dependence of the polynomials P_1, P_2, P_3 with respect to $P_0(x_1, x_2) = x_1$ is the 2-out-of-3 threshold access structure since the polynomial P_0 algebraically depends on every subset of two polynomials in $\{P_1, P_2, P_3\}$. △

Characterization of the Access Structure of Ideal Polynomial Secret Sharing Schemes. Recall that ports of linearly representable matroids admit ideal secret sharing schemes, and the access structure of an ideal linear scheme is the port of the associated linear matroid [30,29]. It is natural to ask if this connection can be generalized to ideal polynomial schemes and algebraic matroids. However, the answer to this question is negative; there are ideal polynomial schemes whose access structures do not coincide with the access structure determined by the algebraic dependence of the sharing polynomials [68,19] (see Example 1.11 and Example 5.7). The main contribution of this work is circumventing these negative results for large fields. We show that if the field is large enough and the polynomial degrees are low, then under some restrictions, the access structure of an ideal polynomial scheme is determined by the algebraic dependence of the polynomials as explained below.

We prove two incomparable results in our work. Our first theorem considers schemes with polynomial sharing (as discussed above) and polynomial reconstruction, i.e., for every authorized set A holding shares $(\mathsf{sh}_i)_{i\in A}$ for the secret s, there is a polynomial Q_A such that $Q_A((\mathsf{sh}_i)_{i\in A}) = s$, i.e., a polynomial that reconstructs the secret.

Theorem 1.4. *Let Σ be an ideal polynomial secret sharing schemes over the field $\mathbb{F}_q$ with sharing and reconstruction degrees at most d_1 and d_2 respectively. If $q > \max\{d_1^{n+1}, d_1 d_2\}$, then the access structure of Σ is determined by the algebraic dependence of the sharing polynomials.*

In the above theorem, there are no restrictions on the characteristic of the field (e.g., it applies to $\mathbb{F}_{2^\ell}$ for large enough ℓ). We show that a restriction on the degree of the polynomials is needed for Theorem 1.4 to hold; there are perfect polynomial schemes (over small fields) that do not realize the access structure determined by the algebraic dependence of the sharing polynomials. This is discussed in Section 5, where we show such perfect polynomial schemes over $\mathbb{F}_q$ with $q = d_1^n$. See also Example 1.11.

In the case that the field is prime, we prove that the access structure of a polynomial scheme is the one determined by the algebraic dependence of the sharing polynomials without any restriction on the degree of the reconstruction polynomials.

Theorem 1.5. *Let Σ be an ideal perfect secret sharing scheme with polynomial sharing in a field of prime order p with sharing polynomials of degree at most d. If $p = \Omega(nd^{8n})$, then the access structure of Σ is determined by the algebraic dependence of the sharing polynomials.*

We can restate the previous theorem in terms of the algebraic matroid defined by the sharing polynomials and the polynomial $P_0(s, r_1, \ldots, r_m) = s$, namely, a matroid in which a set A is dependent if and only if the polynomials $\{P_i\}_{i\in A}$ are algebraically dependent. The following corollary is a new connection between ideal schemes and matroids, which generalizes the connection between ideal linear schemes and representable matroids to the algebraic context.

Corollary 1.6. *Let Σ be an ideal secret sharing scheme satisfying the conditions of Theorem 1.4 or Theorem 1.5, then the access structure of Σ is a port of the algebraic matroid represented algebraically by the sharing polynomials.*

Constructing Secret Sharing Schemes from Polynomials. We next discuss the converse direction, considering the general case where each party may have more than one polynomial, that is, the i-th party has a set of k_i polynomials $\{P_{i,j}\}_{1\leq j\leq k_i}$. Define the access structure Γ as the family of subsets A satisfying that $P_0(s, r_1, \ldots, r_m) = s$ is dependent on $\cup_{i\in A}\{P_{i,1}, \ldots, P_{i,k_i}\}$. Then the question is: Can Γ be realized by a secret sharing scheme with small shares?

We give positive answers to this question, presenting three different constructions. First, we consider the polynomial scheme whose sharing polynomials are $\{P_{i,j}\}_{1\leq j\leq k_i}$. Namely, the shares for a secret s are computed by uniformly taking $r_1, \ldots, r_m \in \mathbb{F}$ and giving the shares $P_{i,1}(s, r_1, \ldots, r_m), \ldots, P_{i,k_i}(s, r_1, \ldots, r_m)$ to the i-th party. As discussed above, these schemes may be non-perfect. We analyze the privacy and correctness properties of these schemes in Theorem 4.8, and show that when the field is prime and large enough, the subsets in Γ have, on average, more information about the secret than the subsets not in Γ. Such schemes are called partial secret sharing schemes [54].

Second, we show that, under some restrictions, there are linear schemes realizing Γ from these polynomials. Namely, we show that if these polynomials are of small degree and are defined over fields with large characteristic, then there is a linear secret sharing scheme with access structure Γ over a extension field.

Theorem 1.7 (Informal). *Let n, m, k, t and d be positive integers and let q be a prime power. For every $1 \leq i \leq n$, let $\{P_{i,j}\}_{1\leq j\leq k_i}$ be some polynomials in $\mathbb{F}_q[x_0, \ldots, x_m]$ of degree at most d, and let $P_0 \in \mathbb{F}_q[x_0, \ldots, x_m]$ be any polynomial. Let r be the algebraic rank of these polynomials. Let Γ be the n-party access structure determined by the polynomials $\{P_{i,j}\}_{i,j}$ and P_0 and let $k = \sum_{i\in[n]} k_i$.*

If $\mathrm{char}(\mathbb{F}_q) > d^r$, then Γ admits a linear secret sharing scheme over $\mathbb{F}_{q^\ell}$ for any $\ell > (r(\log \frac{e(k+1)}{r}) + \log dr)/\log q$ with secret of size $\ell \log q$ and total share size $k\ell \log q$ (i.e., information ratio k).

This result is important for understanding the power of polynomial schemes. It implies that polynomial schemes can provide a meaningful efficiency gain with respect to the linear ones only for fields of small characteristic. This is the case of the polynomial schemes in [17], where the fields have characteristic 2.

The third construction explores another family of schemes that was implicitly introduced in [69]. These schemes, as explained later, take points in an algebraic variety defined by the annihilators of the polynomials. As in the first construction, we can obtain partial schemes where the subsets in Γ have more information about the secret (on average) than the subsets not in Γ. In this case, we do not have any restriction on the characteristic of the field. However, we do not have a bound on the secret size of this partial scheme.

These schemes, using a result of [54], can be turned into statistical schemes, providing more security guarantees. A statistical scheme with security parameter ℓ satisfies the following: Authorized subsets may fail to reconstruct the secret with a negligible probability of error in ℓ, and unauthorized subsets may obtain some amount of information about the secret that is negligible in ℓ.

Theorem 1.8 (Informal). *Let n, k be positive integers and let $\mathbb{F}$ be a finite field. Assume that each party $1 \leq i \leq n$ has k_i polynomials over $\mathbb{F}$, and there are k polynomials in total. Let Γ be the access structure determined by these polynomials as above. Then there exists a statistical scheme realizing Γ with information ratio that tends to $\max_i k_i$ as the security parameter increases.*

To prove this result, we first consider the case in which each party has only one polynomial. For this case, we show that this scheme exists by applying a result of matroid theory. Namely, we prove the following result and then extend it to the general case.

Theorem 1.9. *Let Γ be a port of an algebraic matroid over a finite field. The access structure Γ admits a statistical secret sharing schemes with information ratio tending to 1 as the security parameter increases.*

As a consequence of this result, we give a negative answer for an open question about secret sharing schemes and duality, in the case of statistical schemes. Namely, the dual of an access structure Γ on a set E is $\Gamma^* = \{A \subseteq E : E \setminus A \notin \Gamma\}$. In general, it is not known if duality preserves the optimal information ratio of an access structure or the optimal share size, even for the ideal case. Combining the results on statistical secret sharing schemes from [54] with the fact that there exist almost entropic matroids whose dual is not almost entropic [36,56], we obtain the following result.

Theorem 1.10. *The optimal information ratio of statistical schemes for an access structure is not preserved by duality.*

Multi-linear Secret Sharing and Polynomial Secret Sharing. Multi-linear schemes generalize the linear ones by considering secrets that are vectors of elements in a finite field [22,24,40]. There is an interesting family of perfect polynomial sharing schemes, the one determined by q-polynomials. These are polynomials over $\mathbb{F}_{q^r}$ where the monomials are powers of a single variable, and the exponent is a power of q. For example, $P_1(x_0, x_1, x_2) = x_0 + x_1 - x_2^q$ and the polynomials described in Example 5.7 are q-polynomials. We show in Theorem 5.2 that every multi-linear scheme can be described as a polynomial scheme whose sharing polynomials are q-polynomials. In particular, every multi-linear secret sharing scheme is a polynomial scheme, where the degree of the sharing polynomials is bounded.

1.3 Our Techniques

We divide this section in two parts. First, we explain the techniques used in Theorem 1.4 and Theorem 1.5 for the characterization of the access structure of ideal polynomial schemes. Then, we present the techniques used in Theorem 1.7 and Theorem 1.8 to construct linear, partial, and statistical schemes.

Characterizing Access Structure of Ideal Polynomial Secret Sharing Schemes. In Theorem 1.4, we prove that, under some conditions, the access structure of an ideal polynomial scheme with polynomial sharing and polynomial reconstruction is indeed the one defined by the algebraic dependence of the polynomials. To prove this theorem, we consider an ideal polynomial secret sharing scheme over $\mathbb{F}_q$ realizing an n-party access structure Γ. Let $P_0(s, r_1, \ldots, r_m) = s$ and let P_i be the polynomial that computes the share of the i-th party for $1 \leq i \leq n$. We want to prove that a set of parties A can reconstruct the secret if and only if the polynomial P_0 algebraically depends on the polynomials $\{P_i\}_{i \in A}$. This implies, by properties of matroids, that the matroid defined by the polynomials $P_0, \ldots, P_n$ is the matroid defined by the ideal scheme.

First, assume that a set of parties A can reconstruct the secret. Since the reconstruction is by polynomials, there exists a polynomial $Q_A(y_1, \ldots, y_{|A|})$ such that $Q_A(\{P_i(s, r_1, \ldots, r_m)\}_{i \in A}) = s$ for every $s, r_1, \ldots, r_m \in \mathbb{F}_q$. Notice that, over finite fields, this does not imply that $Q_A(\{P_i\}_{i \in A}) = P_0$, as there are non-zero polynomials that evaluate to zero on all the points (e.g., the polynomial $x^q - x$ over the field $\mathbb{F}_q$).

Let R be the polynomial $R(s, r_1, \ldots, r_m) = Q_A(\{P_i(s, r_1, \ldots, r_m)\}_{i \in A}) - P_0(s, r_1, \ldots, r_m)$. According to the conditions of Theorem 1.4, the degree of the sharing polynomials is at most d_1 and the degree of the reconstruction polynomial is at most d_2; hence, the degree of $R(s, r_1, \ldots, r_m)$ is at most $d_1 \cdot d_2 < q$. By the Schwartz–Zippel Lemma (also called DeMillo–Lipton–Schwartz–Zippel Lemma), any non-zero polynomial with degree less than q is not identically zero. Thus, $R(s, r_1, \ldots, r_m)$ is the zero polynomial, i.e., P_0 algebraically depends on $\{P_i\}_{i \in A}$.

Second, assume that P_0 depends algebraically on $\{P_i\}_{i \in A}$; without loss of generality, assume that A is a minimal set satisfying this property (in particular, $\{P_i\}_{i \in A}$ are independent). In other words, there exists a non-zero polynomial $F(y_0, y_1, \ldots, y_{|A|})$ such that $F(s, \{P_i\}_{i \in A}) = 0$; i.e., F is an annihilator polynomial. By Beecken, Mittman, and Saxena [9], there exists such an annihilator polynomial of degree at most d_1^n. We show that this implies that there is a secret s and shares $(\mathsf{sh}_i)_{i \in A}$ that have positive probability for the secret s and have probability 0 for the secret 0. Therefore, the parties in A have some partial information on the secret. Since each set of parties is either forbidden (i.e., has no information on the secret) or authorized (i.e., can reconstruct the secret), the set A must be authorized and the theorem follows.

In Theorem 1.5, we prove a similar result without requiring polynomial reconstruction. However, the result requires that the scheme is defined over a large prime field. This is because we used the results of Dvir, Gabizon, and Wigderson [42] on rank extractors for polynomial sources. Namely, we use the property that, for large enough prime fields, the output of algebraically independent polynomials is close to having the min-entropy of uniformly distributed random variables. This allows to connect the information theoretic property of the ideal scheme to an algebraic relation between the sharing polynomials.

It is worth noticing that, for fields of large enough characteristic, the algebraic rank of a set of polynomials can be efficiently computed by means of the Jacobian rank of the polynomials [9], which is computed from their partial derivations. Under the conditions of Theorem 1.4, the characteristic is large enough and this property holds. This simplifies the search for authorized subsets given a set of polynomials.

We observe that the bound on the degree of the sharing and reconstruction polynomials in Theorem 1.4, i.e., $q > \max\{d_1^{n+1}, d_1 d_2\}$, is supported by the fact that there exist multi-linear matroids that are not algebraically representable [19]. The scheme determined by one of these multi-linear matroids $\mathcal{M}$ is an ideal multi-linear scheme whose access structure is a port of $\mathcal{M}$. Multi-linear schemes over a field $\mathbb{F}_{q^\ell}$ are actually schemes with polynomial sharing with sharing and reconstruction degrees at most $q^{\ell-1}$ (see Remark 2.12). By [19], the access structure of this polynomial scheme is a port of a matroid that is not algebraic. In particular, the algebraic matroid determined by the algebraic dependence on these polynomials is not $\mathcal{M}$.

Moreover, there are cases where the matroid is algebraic but the algebraic representation does not define the access structure of the scheme; in Example 1.11 we provide an example of a polynomial scheme where the restriction on the degree is not satisfied, and the access structure of the scheme is not the one determined by the algebraic dependence of the polynomials.

Example 1.11. Consider the polynomial scheme with two parties defined by the following polynomials over $\mathbb{F}_4$: $P_0(x_0, x_1, x_2) = x_0$, $P_1(x_0, x_1, x_2) = x_1 + x_2^2$, and $P_2(x_0, x_1, x_2) = x_0 + x_1^2 + x_2$. Each party does not have any information about the secret (as x_1 and x_2 mask the polynomials P_1 and P_2 respectively), and the two parties together can reconstruct the secret using the polynomial $Q(y_1, y_2) = y_1^2 + y_2$. Notice that for every $(s, r_1, r_2) \in \mathbb{F}_4^3$,

$$Q(P_1(s, r_1, r_2), P_2(s, r_1, r_2)) = (r_1 + r_2^2)^2 + s + r_1^2 + r_2 = s + r_2 + r_2^4,$$

and it is equal to r_0 because $r_2 + r_2^4 = 0$ for every $r_2 \in \mathbb{F}_4$. Hence, it is an ideal secret sharing scheme.

Nevertheless, if we analyze the access structure determined by the algebraic dependence of the polynomials, we get that the set of the two parties is unauthorized since the polynomial P_0 does not algebraically depend on the polynomials P_1 and P_2. To prove it we can apply the Jacobian criteria (see Example 5.7 and Section 4.1). Thus, the access structure of the scheme and the one defined by the algebraic dependence do not coincide. In fact, the algebraic matroid defined by P_0, P_1, and P_2 is the uniform matroid with rank 3. $\triangle$

In Example 5.7 we generalize the previous one by giving a scheme on n parties with sharing polynomials over $\mathbb{F}_q$, where $q = p^n$ for some prime p, of degree $d_1 = p$ and reconstruction polynomials of degree $d_2 = p^{n-1}$; this scheme does not realize the access structure determined by the algebraic dependence. In this case, $q = d_1^n = d_1 d_2$. This justifies the bound on the degree of the sharing and reconstruction polynomials of Theorem 1.4.

Secret Sharing Schemes from Polynomials. As discussed above, in the second part of this work we are interested in constructing schemes with small shares for the access structure determined by the algebraic dependence of polynomials. Recall that this access structure Γ is defined as the family of subsets A that satisfy that P_0 depends on $\cup_{i\in A}\{P_{i,1},\ldots,P_{i,k_i}\}$. For that, we use different techniques that combine some classic results about polynomials over finite fields and more modern results about algebraic matroids.

First, we consider the straightforward option of evaluating the polynomials. That is, we consider the scheme where the i-th party receives $P_{i,1}(s,r_1,\ldots,r_m)$, $\ldots,P_{i,k_i}(s,r_1,\ldots,r_m)$ as shares for the secret s for uniformly distributed random elements $r_1,\ldots,r_m\in\mathbb{F}_q$. We observe that the amount of information of sets of parties is related to the algebraic dependence of P_0 with respect to the polynomials they are given. We show that when the field is prime and large enough, the subsets in Γ can nearly determine the secret, on average; namely, we show that the entropy of the secret given the shares of subsets in Γ is very low. In contrast, the subsets not in Γ have on average almost no information about the secret. The schemes, like the ones we get, where authorized subsets always have more information about the secret than those that are not authorized, are called *partial* secret sharing schemes [54]. We prove the privacy and correctness guarantees by counting the number of possible outputs of the mappings following [42]; we bound this number by using Wooley's Theorem [79] and the Schwartz-Zippel Lemma [39].

Partial secret sharing schemes do not provide the security required for many cryptographic applications. However, they can be transformed into statistical schemes using the results of [54]. Moreover, we have found ramp schemes whose trade-off between share size and the gap is almost optimal (see Example 4.16).

Our second construction, stated in Theorem 1.7, considers polynomials over a finite field $\mathbb{F}_q$ with large characteristic, where q is not necessarily a prime. The key tool in this approach is the use of the Jacobian matrix to compute the algebraic rank of a set of polynomials [9,42]. Using the linearization technique of [2], we show that if we evaluate the Jacobian matrix of P_0 and $\{P_{i,j}\}_{i,j}$ on some elements of an extension field $\mathbb{F}_{q^\ell}$ then the resulting matrix determines a linear secret sharing scheme realizing the access structure determined by the polynomials. This is done by proving that the Jacobian matrix evaluated at these elements satisfies that the Jacobian rank of the polynomials in maximal independent sets I coincides with the rank of the rows in the matrix labeled by I. We show that this result implies that polynomial secret sharing schemes can be more efficient than linear ones when the characteristic is small.

This result also has implications for ideal polynomial schemes. Under the same restrictions on the characteristic, and if the field is large enough, we show that if $\{P_0,\ldots,P_n\}$ defines an ideal polynomial scheme, then there is an ideal linear scheme over a field extension $\mathbb{F}_{q^\ell}$. This is proven by combining Theorem 1.4 and Theorem 1.7.

Statistical Secret Sharing Schemes from Algebraic Varieties. In Theorem 1.8, we construct asymptotically ideal statistical schemes for a port of a matroid Γ without any restrictions on the characteristic of the field. However, in this scheme, we cannot provide bounds on the secret size.

For this result, we introduce a new family of secret sharing schemes whose shares are determined by points on an algebraic variety. The study of these schemes is motivated by recent results of Matúš [69]. In the following, we present these schemes as a generalization of ideal linear schemes. In an ideal linear scheme, the i-th party is associated with a vector $v_i \in \mathbb{F}^m$. The shares of the i-th party are $v_i \cdot x$ for some $x \in \mathbb{F}^m$. If we consider the matrix whose columns are the vectors v_i, we get the generator matrix $C = (v_0, \dots, v_n)$ of a linear code whose codewords are the shares of the scheme. Analogously, we can define the family of vectors of shares of the scheme using the dual code $C^\perp$; a vector $y = (y_0 \dots, y_n) \in \mathbb{F}^{n+1}$ is a vector of shares of the scheme if and only if $yC^\perp = 0$. To share a secret s, it suffices to pick uniformly at random vector in the kernel of $C^\perp$ with $y_0 = s$.

Our construction generalizes the scheme described by the dual code as follows. Recall that an annihilator of polynomials $P_1, \dots, P_k \in \mathbb{F}[x_0, \dots, x_m]$ is a polynomial $Q \in \mathbb{F}[y_1, \dots, y_k]$ that satisfies $Q(P_1, \dots, P_k) = 0$. Given a set of polynomials $P_0, \dots, P_n$, we consider the family of their annihilators I, which plays the role of the dual code in the scheme presented above. Then, we consider the family of points in $\mathbb{F}^{n+1}$ that are zeros of all polynomials in I. That is, we consider the points $(y_0, \dots, y_n) \in \mathbb{F}^{n+1}$ such that $Q(y_0, \dots, y_n) = 0$ for every $Q \in I$. In the context of algebraic geometry, this family of points is called the algebraic variety determined by I and is denoted by $V(I)$. In our scheme, we pick a point in the algebraic variety uniformly at random. The resulting scheme is not a scheme with polynomial sharing, in general. It can differ from the scheme whose sharing polynomials are $P_0, \dots, P_n$; in particular, the number of points in $V(I)$ may be higher than the number of possible outputs of the polynomials in I.

In [69], Matúš defined sequences of algebraic varieties defined over extensions of $\mathbb{F}$. That is, the families of points in extensions $\mathbb{K}$ of $\mathbb{F}$ that are zeros of all polynomials in I. Matúš studied the class of almost entropic matroids by considering random variables that are obtained by taking uniform distributions over these varieties. He proved that algebraic matroids are almost entropic (see Section 6 for more details). The proof of these results uses the Lang-Weil bound [61] to bound the number of points in each of these algebraic varieties.

We adapt these results to show that, taking a large enough extension of the field, these random variables define partial secret sharing schemes for the access structure Γ determined by $P_0, \dots, P_n$. Then, we extend this result to the non-ideal case where each party has more than one polynomial.

Partial secret sharing schemes do not provide the security required for many cryptographic applications, so we transform them into schemes with statistical security. We apply a black-box transformation of Jafari and Khazaei [54] that is based on wiretap channel techniques [37,80,81]. They showed that, given a sequence of partial schemes whose privacy and correctness are perfect in the limit and their information ratio converges to a constant, it is possible to create a statistical secret sharing scheme whose information ratio tends to that constant by increasing the security parameter.

As a side result, we observe that the optimal information ratio of statistical schemes is not preserved by duality, proving Theorem 1.10. This is done by combining the results of statistical schemes of [54] applied to access structures defined by the algebraic dependence of polynomials (Theorem 1.8) with some previous results on duality properties of matroids [36,56].

1.4 Related Work

Ideal Secret Sharing Schemes and Matroids. Brickell and Davenport [30] proved that the access structure of an ideal secret sharing scheme determines a matroid, and that the access structure is indeed a port of this matroid. In contrast, only the ports of certain matroids admit ideal schemes, and these matroids are called *entropic* [68,77]. Matroids that admit linear or multi-linear representations are entropic. Seymour [75] proved that there is an access structure that is a port of a matroid and does not have an ideal scheme. Matúš showed that there exist algebraic matroids that are not entropic [68], and Ben-Efraim showed that there exist entropic matroids that are not algebraic [19]. The characterization of ideal access structures is still an open problem.

The connection between ideal schemes and matroids provides mathematical tools for the construction of ideal schemes and, among particular families of access structures, to characterize the ones that admit ideal schemes (e.g. [12,18,45,46]). Farràs [43] proved that almost all matroid ports require linear schemes with share size exponential in the number of parties. However, the best lower bounds on the information ratio for general schemes realizing matroid ports are just constant [7,51]. Martí-Farré and Padró [66] showed that for access structures that are not matroid ports, the information ratio of any secret sharing scheme is at least 3/2.

Polynomial Secret Sharing Schemes. Liu, Vaikuntanathan, and Wee [65] constructed two-server Conditional Disclosure of Secrets (CDS) protocols, where the sharing and reconstruction functions are quadratic polynomials. CDS protocols are basically equivalent to secret sharing schemes for the so-called forbidden access structures. Paskin-Cherniavsky and Radune [72] defined polynomial secret sharing schemes, studied the randomness complexity of these schemes, and gave an exponential upper bound on the randomness complexity. Beimel, Othman, and Peter [17] studied the family of schemes and conditional disclosure of secrets protocols with low degree polynomial reconstruction, finding lower bounds on the share size. They also show that, under plausible assumptions, secret sharing schemes with polynomial sharing are more efficient than secret sharing schemes with polynomial reconstruction. They provided constructions of secret sharing schemes with quadratic sharing and reconstruction, i.e., by polynomials of degree two, whose share size improve the best upper bound on the share size for linear schemes. This line of work was continued in [14], providing secret sharing schemes with polynomial reconstruction of degree d for general access structures, with share size decreasing with d.

Polynomial Sources and Rank Extractors. Dvir, Gabizon, and Wigderson introduced polynomial sources [42], a class of distributions obtained by evaluating some low-degree polynomials on a uniform input. They constructed deterministic randomness extractors using polynomial sources, generalizing previous extractors from affine sources. In particular, they proved that algebraic independence is related to the min-entropy of the polynomial sources. For that, they used a relation between the rank of the polynomials and the rank of the corresponding Jacobian matrix and Wooley's theorem to get a lower bound on the entropy of a polynomial source. Later, Dvir [41] extended this work by considering sources that are distributed uniformly on an algebraic variety. In a different context, Matúš also studied the entropy of these random variables [69]. Beecken, Mittman, and Sexena [9] gave a bound on the degree of the annihilator polynomial of some set of dependent polynomials, improving previous bounds in [42,59]. In this work, we use the well-known Schwartz-Zippel lemma bounding the number of zeros of a polynomial in terms of its degree. We also use results in [9,42,69] to approximate the entropy of polynomial sharings, and our generalization from linear to polynomial schemes is analogous to their generalization from affine to polynomial sources. Moreover, we apply the linearization techniques by Applebaum, Avron, and Brzuska presented in [2]. These techniques were developed to solve the so-called arithmetic predictability problem.

1.5 Discussion

Upper and Lower Bounds for Polynomial Schemes. For the class of linear secret sharing schemes, the best upper bound on the share size is $2^{0.7563n}$ [1,3,4,5,64], and the best lower bound is $2^{0.5n}$ [6]. For schemes with polynomial reconstruction, there are upper and lower bounds on the share size that depend on the degree of polynomials [14,17]. The best upper bound for secret sharing schemes with quadratic sharing is $2^{0.705n}$ [17]; it is not known how to improve the share size taking higher degree sharing polynomials. The best known lower bound for polynomial secret sharing schemes is $\Omega(n/\log n)$ [35], which holds for arbitrary secret sharing schemes. The fact that the randomness of polynomials can be arbitrarily large hinders the use of counting arguments to prove lower bounds on the share size.

Schemes from Algebraic Varieties. When proving Theorem 1.8, we worked with a class of secret sharing schemes whose shares are the points of an algebraic variety. This construction is implicitly used in [69] in a different context. These schemes generalize the linear ones because the shares of linear schemes are points in an algebraic variety determined by polynomials of degree one. We have seen that, increasing the degree of these polynomials, we get more expressivity, being able to get statistical schemes with information ratio close to one for access structures that do not admit ideal linear schemes. Recently, the construction of Matúš has been generalized to definable sets in first-order languages [25].

Potential Applications. Besides the interest of polynomial schemes for reducing the share size, the fact that they are not linear could be useful for some applications. There are other related cryptographic primitives that make use of polynomial mappings and algebraic varieties to guarantee non-malleability or leakage-resilience in different contexts. For example, there are algebraic manipulation detection (AMD) codes whose codewords are points in algebraic varieties (e.g. [28]), AMD codes defined by polynomial mappings (e.g. [34]), and leakage-resilient schemes based on the vector inner-product (e.g. [38]). Therefore, it is worth exploring the applicability of polynomial secret sharing schemes in these contexts.

Construction of Statistical Schemes. The statistical secret sharing schemes in Theorem 1.8 are built with a generic transformation from partial secret sharing schemes to statistical schemes [54]. These statistical schemes could possibly be improved by using specific properties of polynomials. In the case of Theorem 1.8, we have no bounds on the size of the share and the secret. We know that the Lang-Weil bound [61] on the number of points of algebraic varieties cannot be significantly improved, in general [78]. However, for specific varieties, these bounds could be improved [31,67].

Duality. Given a linear scheme for an access structure, it is possible to build a linear scheme with the same share size for the dual access structure [50,47,44]. It is not known whether this property can be extended to secret sharing schemes with polynomial sharing or reconstruction, or to secret sharing schemes in general. Similarly, it is not known whether the class of algebraic matroids and the class of entropic matroids are closed by duality. In Theorem 1.10, it is shown that for some access structures there is a small constant gap between the optimal information ratio of statistical secret sharing schemes for an access structure and its dual. However, it is still not known if the optimal information ratio of perfect secret sharing schemes is closed by duality. Indeed, we do not know what the greatest gap is for the share size of an access structure and its dual.

1.6 Organization

In Section 2, we present preliminaries on secret sharing schemes, matroids, and polynomials over finite fields. In Section 3, we prove the first results of this work, Theorem 1.4 and Theorem 1.5. Then, in Section 4, we prove Theorem 1.7. In Section 5 we introduce the family of q-polynomials. In Section 6, we prove Theorem 1.8 and Theorem 1.9. Finally, in Section 6.4 we prove Theorem 1.10. In the appendix, we present supplementary material, including background definitions, proofs, technical details, and observations.

2 Preliminaries

We present the preliminaries of this work in three subsections. The first one defines secret sharing schemes, the second one discusses matroids and their connections to secret sharing schemes, and the third one provides some results of polynomials.

Notation. We denote the power set of a set E by 2^E. The statistical distance between random variables X and Y is defined as $\mathsf{SD}(X,Y) = \frac{1}{2}\sum_x |\mathsf{Pr}(X = x) - \mathsf{Pr}(Y = x)|$, and the Shannon entropy of a random variable X is defined as $H(X) = \sum_x \mathsf{Pr}(X = x) \log 1/\mathsf{Pr}(X = x)$, where the sum is taken over the support of X.

2.1 Secret Sharing Schemes

In this work, we deal with secret sharing schemes with different security notions. In this section, we present the definition of perfect secret sharing schemes (from [10]). Afterwards, we define polynomial secret sharing schemes. Later in this work, in Section 6.1, we give another definition of security for the schemes, statistic security (from [16]). Other notions needed for intermediate results, are provided in Appendix D.

Definition 2.1. *For a set of n parties, a family Γ of subsets of $\{1,\ldots,n\}$ is called monotone if $A \in \Gamma$ and $A \subseteq B$ implies $B \in \Gamma$. An* access structure *is a monotone collection of nonempty subsets of $\{1,\ldots,n\}$. The sets in Γ are called* authorized *and the sets not in Γ are called* forbidden.

Definition 2.2 (Secret Sharing Schemes). *Let K be a finite set of secrets, $|K| \geq 2$. A* secret sharing scheme *is a pair $\Sigma = \langle \Pi, \mu \rangle$, where μ is a probability distribution over some finite set R, and Π is a mapping from $K \times R$ to a set of n-tuples $K_1 \times \cdots \times K_n$, where K_i is called the share-domain of party i. A dealer distributes a secret $s \in K$ by first computing a vector of shares $\Pi(s;r) = (\mathsf{sh}_1,\ldots,\mathsf{sh}_n)$, and then giving sh_i to the party i. For every $A \subseteq [n]$, let $\Pi_A(s;r)$ be the restriction of $\Pi(s;r)$ to its A-entries. We say that the scheme $\Sigma = \langle \Pi, \mu \rangle$ is a secret sharing scheme realizing an access structure Γ if the following two requirements hold:*

Perfect Correctness: *For any authorized set $A \in \Gamma$, there exists a reconstruction function $\textsc{Recon}_A$ such that for every secret $s \in K$ and $r \in R$,*

$$\textsc{Recon}_A(\Pi_A(s;r)) = s).$$

Perfect Privacy: *For any forbidden set $B \notin \Gamma$, for every two secrets $a, b \in S$, and for every possible vector of shares $\langle \mathsf{sh}_j \rangle_{j \in B}$,*

$$\mathsf{Pr}(\Pi_B(a;r) = \langle \mathsf{sh}_j \rangle_{j \in B}) = \mathsf{Pr}(\Pi_B(b;r) = \langle \mathsf{sh}_j \rangle_{j \in B}).$$

We define the the share size *of party j as* $\log |K_j|$, *the* share size *of the scheme as* $\max_{1 \le j \le n}\{\log |K_j|\}$, *and the* information ratio *of the scheme as*

$$\sigma(\Sigma) = \frac{\max_{1 \le i \le n} \log |K_i|}{\log |K|}.$$

Definition 2.3 (Ideal Secret Sharing Schemes and Ideal Access Structures). *We say that a secret sharing scheme Σ is* ideal *if $\sigma(\Sigma) = 1$, i.e., the size of the domain of shares of each party is the size of the domain of secrets. We say that an access structure is ideal if there is an ideal secret sharing scheme realizing it (with some finite domain of secrets).*

When designing ideal secret sharing schemes, we would like the domain of secrets (and shares) to be as small as possible.

Next, we define the schemes that are the subject of the study of this work, polynomial secret sharing schemes.

Definition 2.4 (Polynomial Secret Sharing Schemes). *Let $\mathbb{F}_q$ be a finite field. A secret sharing scheme with* degree-d polynomial sharing $\Sigma = \langle \Pi, \mu \rangle$ *is a secret sharing scheme in which $S = \mathbb{F}_q$ is the domain of secrets, $R = \mathbb{F}_q^m$ is the randomness space for some $m > 0$, μ is the uniform distribution, and for $1 \le i \le n$, the i-th party's share is*

$$\Pi(s;r)_i = ((P_{i,1}(s,r), \ldots, P_{i,k_i}(s,r))$$

for some $k_i > 0$, where each $P_{i,j}(s,r)$ is a degree-d multivariate polynomial in $\mathbb{F}_q[x_0, x_1, \ldots, x_m]$. A secret sharing scheme $\Sigma = \langle \Pi, \mu \rangle$ with degree-d polynomial reconstruction *is a secret sharing scheme in which for every $1 \le i \le n$ the share of the i-th party is a vector $\mathsf{sh}_i = (\mathsf{sh}_{i,1}, \ldots, \mathsf{sh}_{i,k_i}) \in \mathbb{F}_q^{k_i}$ for some $k_i \in \mathbb{N}$ and for every authorized set A, the reconstruction function $\textsc{Recon}_A$ is a polynomial of degree at most d that outputs s when evaluating on the shares $(\mathsf{sh}_{i,j})_{i \in A, 1 \le j \le k_i}$.*

Beimel et al. [14] define polynomial schemes where the set of secrets can be some secret in $S \subseteq \mathbb{F}$ and Paskin-Cherniavsky and Radune [72] consider the case where $= \mathbb{F}^k$ for some $k \ge 1$. We only consider $S = \mathbb{F}$. If $S = \mathbb{F}$ and the sharing polynomials are of degree 1 (or, equivalently, the reconstruction polynomials are of degree 1), then the scheme is *linear*.

In Section 5, we establish a connection between polynomial schemes and a family of schemes that generalize the linear ones and are called *multi-linear* or *k-linear*. In a *k-linear* secret sharing scheme, the domain of the secrets is $S = \mathbb{F}^k$ and the sharing functions are linear mappings. See [11] for more details.

2.2 Matroids

In this section, we first present the definition of matroids and of algebraic matroids, which are the matroids we are focused on. For an introduction to matroid theory, see e.g. [71]. We then present the relation between matroids and ideal secret sharing schemes.

Definition 2.5 (Matroid). *Given a finite set E and a function $r\colon 2^E \to \mathbb{Z}$, the pair (E, r) is called a* matroid *if the following axioms are satisfied for all $X, Y \subseteq E$.*

1. $0 \leq r(X) \leq |X|$.
2. $r(X) \leq r(Y)$ *if* $X \subseteq Y$.
3. $r(X \cap Y) + r(X \cup Y) \leq r(X) + r(Y)$.

The set E and the function r are, respectively, the ground set *and the* rank function *of the matroid.*

Let $\mathcal{M} = (E, r)$ be a matroid. The *independent sets* of $\mathcal{M}$ are the sets $X \subseteq E$ with $r(X) = |X|$. Every subset of an independent set is independent and the maximal independent sets are called *bases*. The *dependent* sets of $\mathcal{M}$ are the sets that are not independent; every set that contains a dependent set is also dependent. The *circuits* of $\mathcal{M}$ are the minimal dependent sets. All bases have the same number of elements, which equals $r(E)$, the *rank of the matroid*. In addition to the definition given in Definition 2.5, there are other equivalent sets of axioms characterizing matroids, which are stated in terms of the properties of the independent sets, circuits, bases, or hyperplanes (see [71]).

Example 2.6. Let $\mathcal{M} = (\{1, 2, 3, 4\}, r)$ be the matroid with rank function

$$r(X) = \min\{|X|, 2\}$$

i.e., the 2-uniform matroid with 4 elements. The independent sets are all sets of size at most 2, the dependent sets are all sets of size at least 3, the bases are the 2-element sets and the circuits are sets with 3 elements. $\triangle$

Definition 2.7 (Ports of Matroids). *Let $\mathcal{M} = (E, r)$ be a matroid and $a \in E$. The port of $\mathcal{M}$ at the point $a \in E$ is a collection of subsets of $E \setminus \{a\}$ defined as*

$$\Gamma = \{A \subseteq E \setminus \{a\} : r(A \cup \{a\}) = r(A)\},$$

i.e., $A \in \Gamma$ if and only if adding a to A does not increase its rank.

A matroid is *connected* if and only if for every two elements in E there is a circuit containing them. A port of a connected matroid Γ determines the matroid $\mathcal{M}$, i.e., if Γ is a port of a connected matroid at some point, then there is a unique matroid satisfying this property (see [71, Prop. 4.1.2 and Th. 4.3.3]).

Algebraic Dependency and Algebraic Matroids. We next briefly recall the notion of algebraic dependency. Let $\mathbb{F}$ be a field and $\mathbb{K}$ be a transcendental extension field of $\mathbb{F}$. An element $a \in \mathbb{K}$ is algebraically dependent on a set $\{a_1, \ldots, a_k\} \subseteq \mathbb{K}$ if and only if there exists a polynomial $F \in \mathbb{F}(y_0, y_1, \ldots, y_k)$ (i.e., with coefficients in $\mathbb{F}$) such that $F(a, a_1, \ldots, a_k) = 0$ and F contains at least one monomial with a non-zero coefficient and with a degree of y_0 greater than 0. Such an F is called an *annihilating polynomial* of $a, a_1, \ldots, a_k$. A set A

in $\mathbb{K}$ is algebraically dependent over $\mathbb{F}$ if and only if there exists $a \in A$ such that a is algebraically dependent on $A \setminus \{a\}$. Conversely, a set B in $\mathbb{K}$ is algebraically independent over $\mathbb{F}$ if it is not dependent. The transcendence degree or algebraic rank of a set A in $\mathbb{K}$, denoted $\mathrm{trdeg}_{\mathbb{F}}(A)$, is the maximal size of a subset $B \subseteq A$ such that B is algebraically independent over $\mathbb{F}$. For every finite set $E \subseteq \mathbb{K}$, the pair $(E, \mathrm{trdeg}_{\mathbb{F}})$ is a matroid (see, e.g., [71]).

Definition 2.8 (Algebraic Matroids). *A matroid $\mathcal{M} = (E, r)$ is* algebraic *over a field $\mathbb{F}$ if there are elements $(a_i)_{i \in E}$ in a field extension $\mathbb{K}$ of $\mathbb{F}$ such that $r(A) = \mathrm{trdeg}_{\mathbb{F}}((a_i)_{i \in A})$ for every set $A \subseteq E$. The elements $(a_i)_{i \in E}$ are called an* algebraic representation *of $\mathcal{M}$.*

See, e.g., [71,63] for more background on algebraic matroids, and [26,52,7,8] for recent results on their characterization. We will mainly consider the case where $\mathbb{K} = \mathbb{F}(x_1, \ldots, x_m)$, i.e., the field of rational functions (of m-variate polynomials) and we will consider representations with m-variant polynomials.

Definition 2.9. *Given polynomials $P_0, \ldots, P_n$, we define the* access structure Γ determined by the polynomials $P_0, \ldots, P_n$ *as the family of subsets $A \subseteq \{1, \ldots, n\}$ such that the polynomial P_0 is algebraically dependent on the set of polynomials $\{P_i\}_{i \in A}$.*

Notice that, for the authorized sets A in Γ, the algebraic rank of $\{P_i\}_{i \in A}$ is the same as the algebraic rank of $\{P_i\}_{i \in A \cup \{0\}}$. Conversely, if the algebraic rank of $\{P_i\}_{i \in A}$ not increase when adding P_0, then the polynomial P_0 is algebraically dependent on $\{P_i\}_{i \in A}$. Therefore, Γ is the port of the algebraic matroid represented by the polynomials $P_0, \ldots, P_n$ at the point 0.

Example 2.10. Let $\mathcal{M} = (\{1, 2, 3\}, r)$ be the 2-uniform matroid with 3 elements, i.e., the matroid in which all sets of size at most 2 are independent. The elements $x_1^2, x_2^2, x_1 + x_2 \in \mathbb{F}_2(x_1, \ldots, x_m)$ are an algebraic representation of $\mathcal{M}$ in $\mathbb{F}_2$, i.e., every two polynomials are independent and the three polynomials are dependent – for $F(y_1, y_2, y_3) = y_1 + y_2 + y_3^2$ we obtain $F(x_1^2, x_2^2, x_1 + x_2) = x_1^2 + x_2^2 + (x_1 + x_2)^2 = 0$ (in $\mathbb{F}_2$). Indeed, the access structure defined by the polynomials $x_1^2, x_2^2, x_1 + x_2$ at any point is the 2-out-of-3 threshold access structure. △

Ideal Secret Sharing Schemes and Matroids. Brickell and Davenport [30] showed that every ideal access structures is a port of a matroid. Formally, given an ideal secret sharing scheme $\Sigma = \langle \Pi, \mu \rangle$ with domain of secrets S, we define $n+1$ random variables $S_0, \ldots, S_n$, obtained by sampling a uniformly distributed secret s in S, choosing $r \in R$, and computing the share $(\mathsf{sh}_1, \ldots, \mathsf{sh}_n) \leftarrow \Pi(s; r)$; S_0 is a random variable representing the secret s and S_i is a random variable representing the share sh_i for $1 \leq i \leq n$. Furthermore, for a set $A \subseteq \{0, \ldots, n\}$ we define $S_A = (S_i)_{i \in A}$.

Theorem 2.11 ([30]). *Given an ideal secret sharing scheme Σ realizing an access structure Γ, define*

$$r(A) = H(S_A)/H(S) = H(S_A)/\log |S|$$

for every $A \subseteq \{0,\ldots,n\}$*. Then,* $\mathcal{M} = (\{0,\ldots,n\}, r)$ *is a matroid and* Γ *is the port of this matroid* $\mathcal{M}$ *at the point* 0*. Furthermore, for every* $A \subseteq \{0,\ldots,n\}$*, the shares* S_A *are uniformly distributed over a domain of size* $|S|^{r(A)}$*.*

The connection between ideal secret sharing schemes and matroids is useful for obtaining results about the characterization of ideal access structures. If an access structure is a port of a linearly representable matroid, then it admits an ideal secret sharing scheme [30]. However, there exist matroids whose ports do not admit ideal schemes [75,68]. Matroids whose ports admit ideal schemes are called *entropic*.

Remark 2.12. We next explain that ideal secret sharing schemes with domain of secrets and randomness space with sizes a power of the same prime number, then they can be seen as polynomial schemes. Recall that every function $f : \mathbb{F}_q^k \to \mathbb{F}_q$ can be represented as a multivariate polynomial (whose degree in each variable is at most $q-1$). Based on the results of [30] (see Theorem 2.11), every ideal secret sharing scheme realizing an access structure Γ whose domain of secrets is S can be obtained as follows: Let Γ be the port of the matroid $\mathcal{M}$ and $B \cup \{0\} \subseteq \{0,\ldots,n\}$ be a basis of $\mathcal{M}$. By [30], the shares of B are uniformly distributed in S^B (independently of the secret) and the share of each $i \in \{1,\ldots,n\} \setminus B$ is determined by the secret and the shares of B (since $r(B\cup\{0\}\cup\{i\}) = r(B\cup\{0\})$). There is a function $f : S \times (\times_{j\in B} S_j) \to S_i$ that computes the share of the i-th party. Thus, in every ideal scheme whose size of the domain of secrets and shares is a prime power q, the shares can be computed by polynomials over $\mathbb{F}_q$.

The observation above motivates the study of ideal access structures and algebraic matroids. As proved by Ben-Efraim [19], there is an ideal access structure whose associated matroid $\mathcal{M}$ is not algebraic. The ideal scheme for this access structure is multi-linear, and the sharings can be computed by multivariate polynomials $P_1,\ldots,P_n$ over some finite field $\mathbb{F}$. Thus, the algebraic matroid with representation $P_0(s,r) = s, P_1,\ldots,P_n$ is not $\mathcal{M}$.

2.3 Polynomials over Finite Fields

We review some basic notions in relation to polynomials over finite fields. For a finite field $\mathbb{F}$ and a polynomial $P \in \mathbb{F}[x_0,\ldots,x_m]$, we denote by $\deg(P)$ the total degree of P. We write $P = 0$ if P is the zero polynomial. Note that there are polynomials over finite fields that are $P \neq 0$ even though $P(x) = 0$ for all $x \in \mathbb{F}$; for example, if $\mathbb{F}$ is a finite field of order prime p and $P(x) = x^p - x$.

Here we present a result which we will use afterwards that bounds the number of roots of a polynomial.

Lemma 2.13 (The Schwartz-Zippel Lemma [82,74,39]). *Let* $\mathbb{F}$ *be a field and let* $f \in \mathbb{F}[x_1,\ldots,x_t]$ *be a non zero polynomial with degree* d*. Then, for any finite subset* $S \subset \mathbb{F}$ *we have*

$$|\{c \in S^t| : f(c) = 0\}| \leq d \cdot |S|^{t-1}.$$

In particular, if $P \in \mathbb{F}_q[x_0,\dots,x_n]$ has degree $< q$ s.t. $P(x_0,\dots,x_n) = 0$ for every $x_0,\dots,x_n \in \mathbb{F}_q^{n+1}$ then $P = 0$.

A bound of the degree of the annihilating polynomial of an algebraically dependent set of polynomials over finite fields is given in [9,55].

Theorem 2.14 ([55, Theorem 4.1.5] and [9, Corollary 5]). *Let $P_1,\dots,P_n \in \mathbb{F}[x_0,\dots,x_m]$ be polynomials of degree at most d and $k = \mathrm{trdeg}_{\mathbb{F}}(P_1,\dots,P_n)$. If $m > k$, then there exists an annihilating polynomial h of $P_1,\dots,P_n$ with $\deg(h) \leq d^k$.*

Recall that the min-entropy of a random variable X is defined as $H_{\min}(X) = \log 1/p_{\max}$, where $p_{\max} = \max_{x\in\mathrm{supp}(X)} \mathsf{Pr}(X = x)$. The following result is a generalization of [42, Theorem 7.8]. See Remark 2.16 for more details.

Theorem 2.15 ([42, Theorem 7.8]). *Let k, m, and d be integers, and $0 < \delta < 1$ be a real number. Let p be a prime such that $p > \max\{(2d)^{\frac{k}{\delta}}, 2^{\frac{10}{\delta}}, (2(2k+1)d^{2k})^{\frac{2}{\delta}}\}$. Let $P_1,\dots,P_n$ be polynomials in $\mathbb{F}_p[x_1,\dots,x_m]$ of degree at most d and let $k = \mathrm{trdeg}_{\mathbb{F}_p}(P_1,\dots,P_n)$. Define the random variable X obtained by evaluating $P_1,\dots,P_n$ over uniformly distributed $x_1,\dots,x_m$ in $\mathbb{F}_p$. Then:*

1. *X has min entropy $\leq (k+\delta)\cdot\log(p)$.*
2. *X is ε-close to having min entropy at least $(k-\delta)\cdot\log(p)$ where $\varepsilon = \frac{2dk}{p}$.*

Remark 2.16. The original theorem in [42] uses n polynomials on n variables and we prove the generalization to an arbitrary number of variables m.

Suppose first that $m < n$, then take $n - m$ additional independent variables and define the polynomials $P'_1,\dots,P'_n$, where $P'_i(x_1,\dots,x_n) = P(x_1,\dots,x_m)$. Since $\mathrm{trdeg}_{\mathbb{F}_p}(P_1,\dots,P_n) = \mathrm{trdeg}_{\mathbb{F}_p}(P'_1,\dots,P'_n)$, the theorem for $m < n$ follows from the theorem for $m = n$.

Now, suppose that $m > n$, and define $P_i(x_1,\dots,x_m) = P_1(x_1,\dots,x_m)$ for $n+1 \leq i \leq m$, i.e., taking $m-n$ copies of the first polynomial, and let Y be the random variable obtained by sampling $x_1,\dots,x_m$ with uniform distribution from $\mathbb{F}_p^m$ and outputting $P_1(x_1,\dots,x_m),\dots,P_m(x_1,\dots,x_m)$. Observe that $\mathrm{trdeg}_{\mathbb{F}_p}(P_1,\dots,P_n) = \mathrm{trdeg}_{\mathbb{F}_p}(P_1,\dots,P_m)$, Furthermore, for every $(a_1,\dots,a_n) \in \mathbb{F}^n$ we have

$$\mathsf{Pr}(X = (a_1,\dots,a_n)) = \mathsf{Pr}(Y = (a_1,\dots,a_n,a_1,\dots,a_1))$$

and for all $(a_1,\dots,a_n,a_{n+1},\dots,a_m)$ such that $a_i \neq a_1$ for some $n+1 \leq i \leq n$, $\mathsf{Pr}(Y = (a_1,\dots,a_m)) = 0$. Hence, the result for $m > n$ follows from the original theorem.

3 Ideal Polynomial Secret Sharing Schemes Determine Algebraic Matroids

Let Γ be an access structure over the set of parties $\{1,\dots,n\}$. Given an ideal secret sharing scheme realizing Γ with polynomial sharing, whose shares are determined by some polynomials $P_1, \dots, P_n$, we consider the algebraic matroid, whose ground set is $\{0, 1, \dots, n\}$ and the algebraic representation is $P_0(s, r) = s$ and $P_1, \dots, P_n$. In this case, the rank of the matroid is determined by the algebraic dependence of the polynomials. On the other hand, as Γ is ideal, it is a port of some matroid. A natural question to ask is if this matroid is the above algebraic matroid.

Question 3.1. Let Γ be an access structure realized by a polynomial ideal secret sharing scheme. Is Γ the port of the algebraic matroid whose algebraic representation is the set of sharing polynomials?

First, we recall that there is a non-algebraic matroid whose ports admit ideal secret sharing schemes [19], and by Remark 2.12, these ideal secret sharing schemes are ideal polynomial schemes. Hence, in general, the access structures realized by ideal polynomial secret sharing schemes are not ports of algebraic matroids. We show that, under some restrictions, the answer to Question 3.1 is positive. We prove two incomparable results. In Theorem 3.2, we show that when the ideal secret sharing scheme has polynomial reconstruction (in addition to the polynomial sharing), the cardinality of the field is big enough, and the degree of the polynomials is small enough, then Γ is indeed a port of this matroid. In Theorem 3.4, we prove a similar result without requiring that the reconstruction is polynomial; however, in this case, we require that the cardinality of the field is prime (i.e., this result does not work for fields whose cardinality is a prime power p^k for $k > 1$). Corollary 1.6 is a direct consequence of Theorem 3.2 and Theorem 3.4.

3.1 Ideal Schemes with Polynomial Sharing and Polynomial Reconstruction

Theorem 3.2 (Theorem 1.4 Restated). *Let Σ be an ideal polynomial secret sharing scheme realizing an n-party access structure Γ. Suppose that the domain of secrets in Σ is $\mathbb{F}_q$ for some prime power q and the sharing and reconstruction are polynomials over $\mathbb{F}_q$ of degree at most d_1 and d_2, respectively. If $q > \max\{d_1^{n+1}, d_1 d_2\}$, then Γ is a port of the algebraic matroid defined by the sharing polynomials of Σ.*

For the proof of this theorem, we need a technical lemma about matroid theory.

Lemma 3.3. *Let $\mathcal{M} = (E, r)$ be a matroid, $0 \in E$, and $A \subseteq E \setminus \{0\}$ be a set such that $r(A) = r(A \cup \{0\})$. Then there is a circuit $C \subseteq A \cup \{0\}$ such that $0 \in C$.*

Proof. Let $B \subseteq A$ be a maximal independent set that is contained in A, that is, an independent set $B \subseteq A$ such that $r(A) = r(B)$. By the monotonicity of the rank and the properties of A and B,

$$r(A \cup \{0\}) = r(A) = r(B) \leq r(B \cup \{0\}) \leq r(A \cup \{0\}).$$

Thus, $r(B) = r(B \cup \{0\})$ and B is an independent set such that $B \cup \{0\}$ is dependent. Thus, there is a circuit $C \subseteq B \cup \{0\}$; since B is independent $0 \in C$. □

Proof of Theorem 3.2. Let $P_1, \dots, P_n \in \mathbb{F}[s, r_1 \dots, r_m]$ be the polynomials that determine the shares in Σ and $P_0(s, r_1, \dots, r_m) = s$. Consider the algebraic matroid $\mathcal{M} = (E, r)$, with $E = \{0, \dots, n\}$ and algebraic representation $\{P_0, \dots, P_n\}$, i.e., for every subset $A \subseteq E$, $r(A) = \mathrm{trdeg}_{\mathbb{F}_q}((P_i)_{i \in A})$. We prove that Γ is the port of the algebraic matroid $\mathcal{M}$ at 0, i.e.,

$$A \text{ is a minimal authorized set in } \Gamma \text{ iff } A \cup \{0\} \text{ is a circuit in } \mathcal{M}. \tag{1}$$

To prove (1), we will prove two claims:

1. If A is a minimal authorized set of Γ, then 0 depends on the set A in $\mathcal{M}$.
2. If $A \cup \{0\}$ is a circuit in $\mathcal{M}$, then A is authorized in Γ.

In the end of the proof, we show that Items 1 and 2 imply (1).

Proving Item 1. Let A be a minimal authorized set in Γ; thus, there exists some polynomial $Q_A \in \mathbb{F}[y_1, \dots, y_{|A|}]$ of degree at most d_2 that reconstructs the secret from the shares of the parties in A, i.e., for every secret $s \in \mathbb{F}_q$ and randomness $r_1, \dots, r_m \in \mathbb{F}_q$,

$$Q_A((P_i(s, r_1, \dots, r_m))_{i \in A}) = s = P_0(s, r_1, \dots, r_m).$$

Consider the polynomial

$$R(s, r_1, \dots, r_m) = Q_A(s, r_1, \dots, r_m) - P_0(s, r_1, \dots, r_m).$$

Observe that $R(s, r_1, \dots, r_m) = 0$ for every assignment $s, r_1, \dots, r_m$. By the hypothesis of the theorem, the degree of the polynomial $R(s, r_1, \dots, r_m)$ is at most $d_1 \cdot d_2 < q$. By the Schwartz–Zippel Lemma (Lemma 2.13), R is identically zero, i.e., the polynomial P_0 is algebraically dependent on $\{P_i\}_{i \in A}$ and $r(A \cup \{0\}) = \mathrm{trdeg}_{\mathbb{F}_q}((P_i)_{i \in A} \cup \{P_0\}) = \mathrm{trdeg}_{\mathbb{F}_q}((P_i)_{i \in A}) = r(A)$.

Proving Item 2. Let $A \cup \{0\}$ be a circuit of $\mathcal{M}$. We prove that A is authorized in Γ. By the perfect privacy of Σ, it suffices to prove that the parties in A get some information on the secret; in particular, we will prove that there exist some shares $(\mathsf{sh}_i)_{i \in A}$ such that have positive probability for a secret s and are not possible for the secret 0.

Since $P_0, \{P_i\}_{i \in A}$ are algebraically dependent, by Theorem 2.14, there exists an annihilating polynomial $F \in \mathbb{F}[y_0, y_1, \dots, y_{|A|}]$ with $\deg(F) \leq d_1^{|A|}$, that is,

$$F(P_0, \{P_i\}_{i \in A}) = 0. \tag{2}$$

Now, consider the polynomial G with variables $s, r_1, \dots, r_m$ defined as

$$G(s, r_1, \dots, r_m) = F(0, (P_i(s, r_1, \dots, r_m))_{i \in A}). \tag{3}$$

Since $\{P_i\}_{i\in A}$ are algebraically independent, the polynomial is not identically zero, i.e., $G(s, r_1, \ldots, r_m) \neq 0$. The degree of G is bounded by the product of degrees

$$\deg(G) \leq \deg(F) \cdot \max\{\deg(P_i)\} \leq (d_1)^{|A|} \cdot d_1 = d_1^{|A|+1} \leq d_1^{n+1} < q.$$

Since the degree of G is less than q, by the Schwartz–Zippel Lemma (Lemma 2.13), there exist some $s, r_1, \ldots, r_m$ for which $G(s, r_1, \ldots, r_m) \neq 0$. Consider the share $\mathsf{sh}_i = P_i(s, r)$ for every $i \in A$. By (2), for any randomness $r'_1, \ldots, r'_m$,

$$F(0, (P_i(0, r'_1, \ldots, r'_m))_{i\in A}) = 0 \quad \text{and} \quad F(0, (\mathsf{sh}_i)_{i\in A}) = G(s, (\mathsf{sh}_i)_{i\in A}) \neq 0.$$

Thus, for any randomness $r' = r'_1, \ldots, r'_m \in \mathbb{F}_q$,

$$(P_i(0, r'))_{i\in A} \neq (\mathsf{sh}_i)_{i\in A}.$$

So the shares $(\mathsf{sh}_i)_{i\in A}$ are impossible given the secret 0 and are possible given the secret s, implying that A is not forbidden, i.e., A is authorized.

We next explain why Items 1 and 2 imply (1). By Item 1, for every minimal authorized set A in Γ, the element 0 depends on the set A, i.e., $r(A) = r(A \cup \{0\})$. By simple properties of matroids (see Lemma 3.3), there is a circuit $C \subseteq A \cup \{0\}$ such that $0 \in C$; let $B = C \setminus \{0\}$. Thus, by Item 2, B is an authorized set. Since $B \subseteq A$ and A is a minimal authorized set, $A = B$ and $A \cup \{0\}$ is a circuit.

By Item 2, for every circuit $A \cup \{0\}$, the set A is an authorized set. If A is not a minimal authorized set, then there is a set $A' \subsetneq A$ such that A' is authorized then. By Item 1, $A' \cup \{0\} \subsetneq A \cup \{0\}$ is dependent, contradicting the fact that $A \cup \{0\}$ is a circuit.

We have proved that $A \cup \{0\}$ is a circuit of the algebraic matroid $\mathcal{M}$ if and only if A is a minimal authorized set of Γ, i.e., Γ is the port of the algebraic matroid. □

3.2 Ideal Schemes with Polynomial Sharing and Arbitrary Reconstruction

The following theorem is proved by using Theorem 2.15.

Theorem 3.4 (Theorem 1.5 Restated). *Let Σ be an ideal polynomial secret sharing scheme realizing an n-party access structure Γ. Suppose that the domain of secrets in Σ is $\mathbb{F}_p$ for some prime p and the sharing is by polynomials over $\mathbb{F}_p$ of degree at most d. If $p > \max\{16(2n+1)^4 d^{8n}, 2^{20}\}$, then Γ is a port of the algebraic matroid defined by the sharing polynomials of Σ.*

Proof. Let $P_1, \ldots, P_n \in \mathbb{F}[s, r_1 \ldots, r_m]$ be the polynomials of degree at most d that determine the shares in Σ and $P_0(s, r_1, \ldots, r_m) = s$. Now, consider the algebraic matroid $\mathcal{M} = (E, r)$, with $E = \{0, \ldots, n\}$ and the algebraic representation $\{P_0, \ldots, P_n\}$, i.e., for every subset $A \subseteq E$, $r(A) = \mathrm{trdeg}_{\mathbb{F}_q}((P_i)_{i\in A})$.

Furthermore, consider the matroid $\mathcal{M}_\Sigma = (E, r_\Sigma)$ defined in Theorem 2.11, where $r_\Sigma(A) = H(S_A)/H(S) = H(S_A)/\log p$. We will prove that these matroids are equal, i.e., $r(A) = r_\Sigma(A)$ for every $A \subseteq \{0, \ldots, n\}$.

Fix a set A and let $k = r(A) = \mathrm{trdeg}_{\mathbb{F}_q}((P_i)_{i\in A})$ and $\delta = 1/2$. Note that the random variable S_A is obtained by sampling $s, r_1, \ldots, r_m$ with uniform distribution and computing the shares as $(P_i(s, r_1, \ldots, r_m))_{i\in A}$, i.e., as in Theorem 2.15. Furthermore,

$$p > \max\{16(2n+1)^4 d^{8n}, 2^{20}\} \geq \max\{(2d)^{2k}, 2^{20}, (2(2k+1)d^{2k})^4\}.$$

Thus, S_A satisfies the hypothesis of Theorem 2.15 with $\delta = 1/2$.

By Theorem 2.15, the random variable S_A has min-entropy at most $(k + 1/2)\log p$ and is ε-close to a random variable Y having min-entropy at least $(k - 1/2)\log p$, with $\varepsilon = 2dk/p$. Furthermore, by Theorem 2.11, the random variable S_A is uniformly distributed over a domain of size $p^{r_\Sigma(A)}$, $H(S_A) = H_{\min}(S_A)$, thus

$$r_\Sigma(A) = \frac{H(S_A)}{\log(p)} = \frac{H_{\min}(S_A)}{\log(p)} \leq k + 0.5.$$

Since $r(A) = k$ and r_Σ are integers, we have that $r_\Sigma(A) \leq r(A)$. For the other direction, assume towards contradiction that $r_\Sigma(A) < r(A)$; as both ranks are integers, $r_\Sigma(A) \leq r(A) - 1$. Thus, S_A is uniformly distributed over a domain D of size at most $p^{r(A)-1}$. Let $p_x = \mathsf{Pr}(S_A = x)$ and $p'_x = \mathsf{Pr}(Y = x)$. By our assumptions, $p_x \geq p^{-(r(A)-1)} > p^{-(r(A)-1/2)} \geq p'_x$ for every $x \in D$. Therefore,

$$2\mathsf{SD}(S_A, Y) \geq \sum_{x\in D} |p_x - p'_x| \geq \sum_{x\in D} p_x - \sum_{x\in D} \frac{1}{p^{r(A)-1/2}} \geq$$
$$\geq 1 - \frac{p^{r(A)-1}}{p^{r(A)-1/2}} = 1 - p^{-1/2}.$$

Thus, since $\mathsf{SD}(S_A, Y) \leq 2dk/p \leq 1/2$,

$$p^{-1/2} \geq 1 - \frac{4dk}{p} \geq \frac{1}{2},$$

contradicting the fact that $p > 2^{20}$. Thus, from the above arguments $r_\Sigma(A) = r(A)$ for every A and the theorem follows. □

4 Secret Sharing Schemes from Polynomials

In the previous section, we proved that, under some conditions, the access structures of ideal polynomial secret sharing schemes are determined by the algebraic rank of the sharing polynomials. It is natural to ask if the converse is true; i.e., given a set of polynomials, can the access structure Γ determined by their algebraic dependence be realized by an ideal scheme? Furthermore, does the scheme defined by the evaluation of such polynomials has valuable properties?

In this section, we give partial answers to these questions. Before presenting our results, in Section 4.1, we give some results on polynomials over finite fields; these results will be required later. Afterwards, in Section 4.2, we show that for fields of large enough characteristic, access structures defined by polynomials can be realized by linear secret sharing schemes. In Section 4.3, we analyze the scheme that is directly obtained from the sharing polynomials. We show that, while this scheme is not perfect in general, when the cardinality of the field is prime and large enough, subsets in Γ can almost determine the secret, on average, and subsets not in Γ have almost no information about the secret.

4.1 Results on Polynomials over Finite Fields

We next quote a criterion for dependence of a set of polynomials by using the partial derivative matrix, or Jacobian. This criterion yields a more efficient way to compute the transcendence degree of polynomials.

Definition 4.1. *For a set of polynomials $P_1, \ldots, P_n \in \mathbb{F}[x_1, \ldots, x_m]$, we define the* Jacobian matrix *(or Jacobian)*[3] *of $P_1, \ldots, P_n$ as the following $n \times m$ matrix:*

$$J_x(P_1, \ldots, P_n) := (\partial_{x_j} P_i)_{i,j} = \begin{pmatrix} \partial_{x_1} P_1 & \ldots & \partial_{x_m} P_1 \\ \vdots & & \vdots \\ \partial_{x_1} P_n & \ldots & \partial_{x_m} P_n \end{pmatrix}$$

with $x = (x_1, \ldots, x_m)$ and $\partial_{x_j} P_i = \frac{\partial P_i}{\partial x_j}$. We define the Jacobian rank *of $P_1, \ldots, P_n$ as the rank of the matrix $J_x(P_1, \ldots, P_n)$ over $\mathbb{F}(x_1, \ldots, x_m)$, and we denote it by $\mathrm{rank}_J(P_1, \ldots, P_n)$.*

Lemma 4.2 (Jacobian criterion [9]). *Let $P_1, \ldots, P_n \in \mathbb{F}[x_1, \ldots, x_m]$ be polynomials of degree at most d and transcendence degree r. If $\mathrm{char}(\mathbb{F}) = 0$ or $\mathrm{char}(\mathbb{F}) > d^r$, then $\mathrm{rank}_J(P_1, \ldots, P_n) = \mathrm{trdeg}_{\mathbb{F}}\{P_1, \ldots, P_n\}$.*

Below, we quote a result of Wooley [79], providing an upper bound on the number of roots of a set of polynomials.

Theorem 4.3 (Wooley's theorem [79]). *Let p be a prime, m and $d_1, \ldots, d_m$ be integers, and $P'_1, \ldots, P'_m$ be polynomials in $\mathbb{F}_p[x_1, \ldots, x_m]$, where the degree of P_i is d_i for $1 \leq i \leq m$. Finally, let*

$$N = |\{c \in \mathbb{F}_p^m \,:\, P_i(c) = 0 \text{ for } 1 \leq i \leq m \text{ and } J_x(P'_1, \ldots, P'_m)(c) \text{ has full rank }\}|$$

Then, $N \leq d_1 \cdot \ldots \cdot d_m$.

In Wooley's theorem, the number of variables and the number of polynomials is the same.[4] We next provide a corollary of Theorem 4.3, considering the case where the number of variables is larger than the number of polynomials. The proof is in Appendix A.

[3] In some places the Jacobian is defined as the determinant of this matrix.

[4] Wooley proves a more general theorem, where the roots are in $\mathbb{Z}_p^{\ell}$ for $\ell \geq 1$. We do not use this generalization.

Theorem 4.4. *Let p be a prime, k, m, and d be integers, $P_1, \dots, P_k$ be polynomials in $\mathbb{F}_p[x_1, \dots, x_m]$ of degree at most d. For $a = (a_1, \dots, a_k) \in \mathbb{F}_p^k$, let*

$$N_a = \left| \left\{ c \in \mathbb{F}_p^m : \begin{array}{c} P_i(c) = a_i \textit{ for } 1 \leq i \leq k \textit{ and} \\ \textit{the first } k \textit{ columns of } J_x(P_1, \dots, P_k)(c) \textit{ are independent} \end{array} \right\} \right| .$$

Then, for every $a \in \mathbb{F}_p^k$, $N_a \leq d^k p^{m-k}$.

4.2 Linear Secret Sharing Schemes from Polynomials

In this subsection, we first present a sufficient condition for an algebraic matroid to be linearly representable. This result is inspired by the techniques developed by Applebaum, Avron, and Brzuska [2] to solve the so-called arithmetic predictability problem. This linear representation of the algebraic matroid implies a linear scheme for the access structure determined by the algebraic dependence of a set of polynomials. In particular, this implies that every ideal polynomial scheme over large prime fields can be converted to a linear scheme for the same access structure.

Theorem 4.5. *Let n, m, and d be positive integers, q a prime power, $P_1, \dots, P_n$ be some polynomials in $\mathbb{F}_q[x_0, \dots, x_m]$ of degree at most d, and $\mathcal{M}$ be the algebraic matroid defined by $P_1, \dots, P_n$. Let r be the rank of $\mathcal{M}$. If $\mathrm{char}(\mathbb{F}_q) > d^r$, then $\mathcal{M}$ is linearly representable over $\mathbb{F}_{q^\ell}$ for every $\ell > (r \log \frac{en}{r} + \log dr)/\log q$.*

Proof. We consider the Jacobian matrix $J_x = J_x(P_1, \dots, P_n)$ of the polynomials $P_0, \dots, P_n$. We will prove that there exists an assignment $a_0, \dots, a_m$ such that $J_x(P_0, \dots, P_n)(a_0, \dots, a_m)$ is a linear representation of $\mathcal{M}$. By Lemma 4.2, for every maximal independent set I in $\mathcal{M}$ there is a minor $J_{I,x}$ of J_x with rows labeled by I and with full rank. Consider the product of all the determinants of the minors obtained by all the maximal independent sets I. Notice that $\det J_{I,x}$ is a polynomial of degree at most dr; therefore, the product of all determinants has degree at most

$$\binom{n}{r} dr \leq \left(\frac{en}{r} \right)^r dr.$$

We consider the field extension $\mathbb{F}_{q^\ell}$ for an ℓ such that $q^\ell > (en/r)^r dr$. By Lemma 2.13, there exists $a_0, \dots, a_m \in \mathbb{F}_{q^\ell}$ such that $\prod_I \det J_{I,x}(a_0, \dots, a_m) \neq 0$. Let

$$J = J_x(P_1, \dots, P_n)(a_0, \dots, a_m),$$

i.e., the matrix obtained from the Jacobian by assigning the values $a_0, \dots, a_m$ to the variables $x_0, \dots, x_m$. We next argue that the matrix J is a linear representation of $\mathcal{M}$. First, for every dependent set A of $\mathcal{M}$ the rank of the rows labeled by A in J is at most the rank of the rows labeled by A in J_x, which is less than $|A|$. Second, for every maximal independent set I of $\mathcal{M}$, by the definition of the above product of determinants and the choice of $a_0, \dots, a_m$, the rows labeled by I in J are independent. $\square$

For a set of parties $\{1,\ldots,n\}$ and a set of polynomials $\{P_{i,j}\}$, for $1 \le i \le n$ and $1 \le j \le k_i$ and P_0, we define the access structure Γ determined by the polynomials P_0 and $\{P_{i,j}\}$ as the family of subsets A satisfying that P_0 is algebraically dependent on $\cup_{i\in A}\{P_{i,1},\ldots,P_{i,k_i}\}$. We show that if the characteristic of $\mathbb{F}_q$ is large enough, the access structure can be realized by a linear scheme over some field extension $\mathbb{F}_{q^\ell}$, .

Corollary 4.6. *Let n, m, k, t, and d be positive integers and let q be a prime power. For every $1 \le i \le n$, let $\{P_{i,j}\}_{1\le j\le k_i}$ be polynomials in $\mathbb{F}_q[x_0,\ldots,x_m]$ of degree at most d, and let $P_0 \in \mathbb{F}_q[x_0,\ldots,x_m]$. Let r be the algebraic rank of these polynomials and let Γ be the n-party access structure determined by the polynomials $\{P_{i,j}\}_{i,j}$ and P_0 and let $k = \sum_{i\in[n]} k_i$.*

If $\mathrm{char}(\mathbb{F}_q) > d^r$, then Γ admits a linear secret sharing scheme over $\mathbb{F}_{q^\ell}$ for any $\ell > (r\log\frac{e(k+1)}{r} + \log dr)/\log q$ with total share size $k\ell\log q$ and size of secrets $\ell\log q$.

Proof. We consider the algebraic matroid on $k = 1+\sum_{i\in[n]} k_i$ elements that is algebraically represented by the polynomials $\{P_{i,j}\}_{1\le j\le k_i}$ and P_0. By Theorem 4.5 there is a linear representation over the field $\mathbb{F}_{q^\ell}$ where $q^\ell > (e(k+1)/r)dr$ where r is the rank of the matroid. By [30], the linear representation induces a linear scheme with total share size $k\ell\log q$. □

In Theorem 1.4 and Theorem 1.5, we proved that, under some conditions, the access structure of an ideal polynomial secret sharing is the one determined by the sharing polynomials. The linear scheme obtained in Corollary 4.6 implies that this access structure also admits a linear scheme. This implies that under some conditions, every ideal polynomial scheme can be converted to an ideal linear scheme.

Corollary 4.7. *Let Σ be an ideal polynomial secret sharing scheme over $\mathbb{F}_q$ with access structure Γ. If one of the following holds:*

(1) The degree of the sharing polynomials is at most d and q is prime with $q = \Omega(nd^{8n})$.
(2) The degree of the sharing and reconstruction polynomials is at most d_1 and d_2, respectively, $q > \max\{d_1^{n+1}, d_1d_2\}$, and $\mathrm{char}(\mathbb{F}_q) > d_1^r$.

then Γ admits an ideal linear secret sharing scheme over a field extension $\mathbb{F}_{q^\ell}$ with $\ell > (r\log\frac{en}{r} + \log dr)/\log q$, where r is the rank of the associated matroid.

4.3 Schemes from Polynomials

In this subsection, we study polynomial schemes over prime fields $\mathbb{F}_p$. We consider the general case where every party may have more than one polynomial and the shares are computed by evaluation polynomials on the secret and random field elements. Inspired by the techniques used in [42], we show that if the field is large enough, this is a partial secret sharing scheme realizing the access structure determined by the algebraic dependence of the polynomials and the polynomial $P_0(s, r_1, \dots, r_m) = s$, that is, the amount of information that an authorized set has about the secret is larger than the information that any forbidden set has about the secret. To prove this result, we give bounds on the entropy of the jointly distributed random variables defined by the polynomials as $S_A = \{P_{i,j}(U_m)\}_{i\in A}$, and we show that the difference between the values of $H(S_0|S_A)$ for authorized and forbidden subsets is positive.

Theorem 4.8. *Let $n, m, k, t,$ and d be positive integers and let p be a prime with $p > d^{t^2+3t+2}$. For every $1 \leq i \leq n$, let $\{P_{i,j}\}_{1\leq j\leq k_i}$ be some polynomials in $\mathbb{F}_p[x_0, \dots, x_m]$ of degree at most d, and $P_0(x_0, \dots, x_m) = x_0$. Let S_A be the random variables defined by the joint distribution $\{P_{i,j}(U_{m+1})\}_{i\in A}$ where U_{m+1} is the uniform distribution over $\mathbb{F}_p^{m+1}$. Let Γ be the n-party access structure determined by the polynomials $\{P_{i,j}\}_{i,j}$ and P_0.*

If $t > \sum_{i\in A} k_i$ for any maximal forbidden subset A not in Γ, then the random variables satisfy that

$$C = \min_{B\notin\Gamma} H(S_0|S_B) - \max_{A\in\Gamma} H(S_0|S_A) \geq \left(1 - \frac{td^t}{p}\right)\log p - (t^2+3t+1)\log d > 0.$$

The proof of this theorem is based on the bounds given in the following lemma.

Lemma 4.9. *Let $n, m, r,$ and d be positive integers and let $p > \max\{d^r, \frac{dn}{2}\}$ be a prime. Let $P_0, \dots, P_n$ be some polynomials in $\mathbb{F}_p[x_0, \dots, x_m]$ of degree at most d and algebraic rank r, with $P_0(x_0, \dots, x_m) = x_0$, and let Γ be the access structure determined by them. For $A \subseteq \{0, \dots, n\}$, let $k = \mathrm{rank}(\{P_i\}_{i\in A})$, and let S_A be the jointly distributed random variable $S_A = \{P_i(U_{m+1})\}_{i\in A}$, where U_{m+1} is the uniform distribution on $\mathbb{F}_p^{m+1}$. Then,*

1. *If A is independent, then $H(S_A) \geq k\log p - k\log 2d$.*
2. *If A is dependent, then $H(S_A) \leq k\log p + (|A| - k)(k\log d + \frac{d^k}{p}\log p)$.*

Proof. By the Jacobian Criterion Lemma 4.2, since $p > d^r$, for every set $A \subseteq \{0, \dots, n\}$, the algebraic rank of $\{P_i\}_{i\in A}$ coincides with the rank of the Jacobian of $\{P_i\}_{i\in A}$.

First, we prove the first statement. Let $k = |A|$. In this case, the rank of the Jacobian is k, i.e., there are k independent columns in the Jacobian, w.l.o.g, assume that these are the first k columns. We apply Theorem 4.4. Let E be the event that given a uniformly distributed assignment c, the first k columns of $J_x(\{P_i\}_{i\in A})(c)$ are linearly independent. By Lemma 2.13, the probability that the determinant of these k columns is zero for a random assignment of $x_0, \dots, x_m$ is at most dk/p.

Now we compute $\mathsf{Pr}(S_A = a \wedge E)$ for $a \in \mathbb{F}_p^k$. By Theorem 4.4, the set N_a is at most $p^{m-k}d^k$, and so $\mathsf{Pr}((S_A = a) \wedge E) \leq d^k/p^k$. Hence, we can bound $\mathsf{Pr}(S_A = a|E)$ as

$$\mathsf{Pr}(S_A = a|E) = \frac{\mathsf{Pr}((S_A = a) \wedge E)}{\mathsf{Pr}(E)} \leq \frac{d^k}{p^k(1 - dk/p)} \leq \left(\frac{2d}{p}\right)^k.$$

This holds since $k \geq 2$ and then $dk < 2p$. Therefore,

$$H(S_A) \geq H(S_A|E) = \sum_{a \in \mathbb{F}_p^k} \mathsf{Pr}(S_A = a|E) \log(1/\mathsf{Pr}(S_A = a|E))$$
$$\geq \sum_{a \in \mathbb{F}_p^k} \mathsf{Pr}(S_A = a|E)(k \log p - k \log(2d)) \geq k \log p - k \log(2d).$$

To prove the second statement, we first consider the case that A is a circuit with rank k. For that, we show the difference between the entropy of S_A and the entropy of every independent subset of A, $S_{A \setminus j}$, for $j \in A$, is bounded by $k \log d + \frac{d^k}{p} \log p$.

Let Q be an annihilator polynomial of A with degree d_Q. By Theorem 2.14, $d_Q \leq d^k$. This polynomial satisfies that for every $x \in \mathbb{F}^m$, $Q(\{P_i(x)\}_{i \in A}) = 0$. Fix $j \in A$ and let $I = A \setminus j$ be an independent set of rank k.

For $a \in \mathbb{F}^{k-1}$ such that $\mathsf{Pr}(S_I = a) > 0$, if $Q(x, a) = 0$ for every $x \in \mathbb{F}_p$, then S_j can take at most q values and; therefore, $H(S_j|S_I = a) \leq \log q$. For $a \in \mathbb{F}^{k-1}$ such that $Q(x, a) \neq 0$ for some x, then there are at most d_Q possible values of x such that $\mathsf{Pr}(S_j = x \wedge S_I = a) > 0$, since all values must satisfy that $Q(x, a) = 0$, a polynomial os x of degree at most d^k. Thus, $H(S_j|S_I = a) \leq \log d_Q$. Summing up,

$$\begin{aligned}
H(S_A) &= H(S_I) + H(S_j|S_I) \\
&= H(S_I) + \sum_{a \in \mathbb{F}^{k-1}} \mathsf{Pr}(S_I = a) H(S_j|S_I = a) \\
&= H(S_I) + \sum_{a \in \mathbb{F}^{k-1}, Q(\cdot, a) = 0} \mathsf{Pr}(S_I = a) H(S_j|S_I = a) \\
&\quad + \sum_{a \in \mathbb{F}^{k-1}, Q(\cdot, a) \neq 0} \mathsf{Pr}(S_I = a) H(S_j|S_I = a) \\
&\leq H(S_I) + \mathsf{Pr}(Q(\cdot, a) = 0) \log q + \mathsf{Pr}(Q(\cdot, a) \neq 0) \log d_Q.
\end{aligned}$$

By Lemma 2.13, the polynomial $Q(\cdot, a)$ is 0 with probability at most $\frac{d_Q}{p}$ since the number of roots of Q is at most $d_Q p^{k-1}$; every such a contributes p roots, i.e., the number of such a's is at most $d_Q p^{k-1}$ and their probability is at most $d_p^{k-1}/p^{k-1} = d_Q/p$. Therefore,

$$\begin{aligned}
H(S_A) - H(S_I) &\leq \frac{d_Q}{p} \log p + \left(1 - \frac{d_Q}{p}\right) \log d_Q = \log d_Q + \frac{d_Q}{p}(\log p - \log d_Q) \\
&\leq k \log d + \frac{d^k}{p}(\log p - \log d_Q) \leq k \log d + \frac{d^k}{p} \log p.
\end{aligned}$$

Finally, we prove the second statement for every dependent subset. Let I be a maximal independent set contained in A. Let $j \in A \setminus I$ and let B_j be the unique circuit contained in $\{j\} \cup I$.

By the submodularity of the entropy function, for all $j \in A \setminus I$,

$$H(S_{B_j}) - H(S_{B_j \setminus j}) \geq H(S_A) - H(S_{A \setminus j}).$$

Therefore,

$$H(S_A) - H(S_I) \leq \sum_{j \in A \setminus I} (H(S_A) - H(S_{A \setminus j})) \leq \sum_{j \in A \setminus I} (H(S_{B_j}) - H(S_{B_j \setminus j})).$$

For every $j \in A \setminus I$, use the bound for the circuit B_j and the independent set $B_j \setminus j$ and the bound for $H(S_I)$, obtaining

$$H(S_A) \leq k \log p + (|A| - k) \left(k \log d + \frac{d^k}{p} \log p \right).$$

□

Theorem 4.8 is implied by the following two claims.

Claim 4.10. *If* $A \in \Gamma$, *then* $H(S_0|S_A) \leq (d^t/p + 3t \log d / \log p) H(S_0)$.

Proof. First, consider the case that A is minimal in Γ. In this case, A is independent and $A \cup \{0\}$ is a circuit. Suppose that $|A| = k$, and so $A \cup \{0\}$ has rank k. By Lemma 4.9, and assuming that the secret is uniformly distributed, i.e., $H(S_0) = \log q$,

$$\begin{aligned} H(S_0|S_A) &\leq k \log p + k \log d + \frac{d^k}{p} \log p - k \log p + k \log 2d \\ &\leq (d^k/p + 3k \log d / \log p) H(S_0). \end{aligned}$$

If $A \in \Gamma$ is not minimal, then there is $A' \subseteq A$ that is minimal in Γ and $H(S_0|S_A) \leq H(S_0|S_{A'})$. Now, since the size of maximal forbidden subsets is smaller than t, the size of minimal authorized subsets is at most t. □

Claim 4.11. *If* $A \notin \Gamma$, *then*

$$H(S_0|S_A) \geq (1 - (t-1)d^{t-1}/p - (t^2+1) \log d / \log p) H(S_0).$$

Proof. Let A be a non authorized set of size $k < t$ and let $k' = \mathrm{rank}(A)$. The rank of $A \cup \{0\}$ is greater than the rank of A, i.e., k'. Therefore, $A \cup \{0\}$ contains an independent set I of size $k'+1$. Thus,

$$\begin{aligned} H(S_0|S_A) &= H(S_{A \cup \{0\}}) - H(S_A) \geq H(S_I) - H(S_A) \\ &\geq (k'+1) \log p - (k'+1) \log 2d - k' \log p - (k - k')(k' \log d + \frac{d^{k'}}{p} \log p) \\ &\geq \log p - (2(k'+1) + (k-k')k') \log d - (k-k') \frac{d^{k'}}{p} \log p \\ &\geq \log p - (k^2 + 2k + 2) \log d - k \frac{d^k}{p} \log p \\ &\geq (1 - kd^k/p - (k^2+2k+2) \log d / \log p) H(S_0). \end{aligned}$$

The claim follows since $t \leq k+1$. □

Therefore the advantage of the authorized sets over the forbidden ones is

$$C = \min_{B \notin \Gamma} H(S_0|S_B) - \max_{A \in \Gamma} H(S_0|S_A) \geq \left(1 - \frac{td^t}{p}\right) \log p - (t^2 + 3t + 1) \log d,$$

which is positive for $p > d^{t^2+3t+2}$ since in this case $p > td^t$.

Remark 4.12. If Γ is a t-out-of-n threshold access structure, we can give a more accurate bound because we know that the maximal forbidden subsets are independent. Therefore, the advantage is

$$C \geq \left(1 - \frac{td^t}{p}\right) \log p - 5t \log d,$$

and we can guarantee that this value is positive for $p > d^{6t}$.

We note that the restriction on the degree of sharing polynomials with respect to the field size is needed. This is justified by Example 6.10, where we give a polynomial scheme not satisfying the restrictions on the degree and for which an unauthorized subset has more information about the secret than an authorized subset.

Remark 4.13. In this theorem we only considered the case that $P_0(x_0, \ldots, x_m) = x_0$. If we consider a scheme where the secret is determined by a polynomial of degree larger than one, it is possible to build a similar scheme with slightly smaller C.

Using Theorem 4.8, we can generalize Theorem 4.5 to the scenario where each party holds more than one polynomial.

Theorem 4.14. *Let P_0 and $\{P_{i,j}\}_{1 \leq j \leq k_i}$ be some polynomials in $\mathbb{F}_p[x_0, \ldots, x_m]$ of degree at most d for every $1 \leq i \leq n$ and $t > \sum_{i \in A} k_i$ for any maximal forbidden subset A. Consider the polynomial scheme with sharing polynomials $\{P_{i,j}\}$ and secret P_0. If the scheme is perfect, with access structure Γ, and if $p > d^{t^2+t}$, then there is a linear scheme over $\mathbb{F}_{p^\ell}$ for Γ for some $\ell > (t \log \frac{en}{t} + \log dt)/\log p$.*

Proof. We will prove that $A \in \Gamma$ iff P_0 algebraically depends on $\{P_{i,j}\}_{i \in A}$. We then construct the linear scheme over $\mathbb{F}_{q^\ell}$ implied by Corollary 4.7.

First, see that if P_0 is not algebraically dependent on A, then by Claim 4.11 $H(S_0|S_A) > (1 - \frac{(t-1)d^{t-1}}{p} + \frac{(t^2+1)\log d}{\log p})H(S_0)$. Since

$$\frac{(t-1)d^{t-1}}{p} + \frac{(t^2+1)\log d}{\log p} < \frac{(t-1)d^{t-1}}{d^{t^2+1}} + \frac{t^2+1}{t^2+t} < 1,$$

then $H(S_0|S_A) > 0$ and by perfect privacy of the scheme, $H(S_0|S_A) = H(S)$.

Next, assume that P_0 is algebraically dependent on $\{P_{i,j}\}_{i\in A\cdot}$, by Claim 4.10 $H(S_0|S_A) < (d^t/p + 3t\log d/\log p)H(S_0)$. Since

$$\frac{d^t}{p} + 3t\frac{\log d}{\log p} < \frac{1}{d^{t^2+t}} + \frac{3t}{t^2+t},$$

then $H(S_0|S_A) < H(S_0)$ and by perfect correctness of the scheme, $H(S_0|S_A) = 0$. □

Remark 4.15. In a more general case where the polynomial scheme is not perfect, we can be also deduce a linear scheme for the access structure partially determined by this scheme. This access structure consists on the collection of subsets that satisfy $H(S_0|S_A) < \frac{1}{2}H(S_0)$.

By Claim 4.11 and Claim 4.10, if $p > d^{2t^2+t}$, a subset A such that P_0 depends algebraically on $\{P_{i,j}\}$ satisfies that $H(S_0|S_A) > \frac{1}{2}H(S_0)$ and a subset A such that P_0 is algebraically independent on $\{P_{i,j}\}$ satisfies that $H(S_0|S_A) < \frac{1}{2}H(S_0)$. Therefore, the linear scheme deduced in Corollary 4.7 is a linear scheme for the access structure of the polynomial scheme.

4.4 Polynomial Ramp Schemes

We show that there are also interesting polynomial schemes. In a *ramp* secret sharing scheme, there are two thresholds $t_1 < t_2$ where subsets of size at least t_2 are authorized (i.e., can always reconstruct the secret) and subsets of size at most t_1 are forbidden (i.e., obtain no information on the secret), t_1 is called the privacy threshold and t_2 is called the reconstruction threshold. Next, we present a polynomial ramp scheme with an almost optimal trade-off between the share size and the gap $t_2 - t_1$.

Example 4.16. Let p be prime. Consider the polynomial scheme defined by the polynomials $P_{a,b}(x,y) = (x-a)(y-b)$ for $a,b \in \mathbb{F}_p$ for the shares and $P_0(x,y) = x$ for the secret. The access structure Γ defined by these polynomials is the 2-out-out-of-n threshold access structure on $n = p^2$ parties and share size $\log p = \frac{1}{2}\log n$.

Indeed, every pair of polynomials P_{a_1,b_1}, P_{a_2,b_2} is algebraically independent over $\mathbb{F}$.

$$J = J_x(P_{a_1,b_1}, P_{a_2,b_2}) = \begin{pmatrix} y-b_1 & x-a_1 \\ y-b_2 & x-a_2 \end{pmatrix}$$

and the determinant is $\det(J) = (y-b_1)(x-a_2) - (y-b_2)(x-a_1) = (b_2 - b_1)x - (a_2 - a_2)t + a_2b_1 - a_1b_2$ so, since $a_1 \neq a_2$ or $b_1 \neq b_2$, this determinant is non-zero. Moreover, every triple of polynomials is algebraically dependent since they are in $\mathbb{F}_q[x,y]$.

Nevertheless, the scheme defined by these polynomials does not realize Γ perfectly. First, notice that there are parties with partial information about the secret. Let $s \in \mathbb{F}_p$, the share of a party associated with a polynomial $P_{s,b}$ is 0 when the secret is s; however, when the secret is $s' \neq s$, the probability that its share is 0 is $1/p$. I.e., the parties have probability $1/p$ to have more information. So the scheme is not 1-private.

Moreover, the correctness for sets with two parties is not perfect, since, in general, the reconstruction function is a degree-2 polynomial and there are cases where the parties do not have any information about the secret. Concretely, let $a_1, a_2, b_1, b_2 \in \mathbb{F}_p$ and two parties with $P_{a_1,b_1,}, P_{a_2,b_2}$ as polynomial shares. If $b_1 = b_2$ and $a_1 \neq a_2$, if $r_1 = b_1$, the parties have the share 0 for any secret and they cannot reconstruct it. Moreover, there are subsets with more than two parties that still do not satisfy correctness for every secret and randomness. For example, the subset of p parties with polynomial share $P_{a_i,b}$, (i.e., with the same b), cannot reconstruct the secret when the randomness $r_1 = b$ since all shares will be 0. The smallest size of subsets that always have perfect correctness is $p + 1$, since all subsets of size $p + 1$ will reconstruct the secret with probability 1. Let $s, r \in \mathbb{F}$, since there are $p + 1$ polynomials, there is at least a pair with same a but different b and then they can reconstruct a secret. This makes the scheme to have $(p + 1)$-reconstruction.

Schemes for p^2 parties over $\mathbb{F}_p$ with share size $\frac{1}{2} \log n$ and privacy threshold 1 require reconstruction threshold at least $p + \frac{1}{p}$ [33, Theorem 3.3]. The reconstruction threshold of our scheme is $p + 1$, which almost achieves the optimal value.

Even tough the scheme does not have good security properties, from the results in Section 4, in particular Remark 4.12, it is deduced that, in average, every authorized subset has more information about the secret that any forbidden one. In particular, let S_0 be the random variable associated to the secret, let S_i be the random variable associated to the share of party i, and let $S_{i,j}$ be the random variable associated to the share of the pair of parties i, j. Then,

$$\min_i H(S_0|S_i) - \max_{i,j} H(S_0|S_{i,j}) \geq \left(1 - \frac{8}{p}\right) \log p - 10,$$

which is very close to $\log p$ for large enough p. △

Notice that for this access structure, by relaxing the privacy and correctness requirements, we could overcome the share size limitations of perfect secret sharing schemes. In the case of 2-threshold access structures, Kilian and Nisan showed that the share size of any perfect scheme sharing for a one-bit secret is at least $\log n$ [60,33]. In our scheme, the number of parties is $n = p^2$, and the share size is $\log p = \frac{1}{2} \log n$.

Looking precisely at the information each share has about the secret, we see that for every selection of $(a, b) \in \mathbb{F}^2$, there is a subset of parties of size p that does not have information about the secret (those parties (a, b) with $b = y$).

Next, we provide a construction based on the Shamir scheme that has similar properties. Compared to the previous one, it has perfect 1-privacy; however it requires public information. The share size of each party is the same, but the amount of public information is the same as the total share size. This variant reduces the known bound of the field for the Shamir scheme but loses perfectness.

Example 4.17. To construct a 2-out-of-n scheme, we first use Shamir's 2-out-of-$\sqrt{n}$ secret sharing scheme [76] over a field of order $p = \sqrt{n}$ and get $\sqrt{n}$ shares $\mathsf{sh}_1, \dots, \mathsf{sh}_{\sqrt{n}}$. We then assign the shares randomly to the n parties, i.e., each party is assigned a random $1 \leq i \leq p$ and the share is sh_i. This assignment is made public. The size of the public information is $n \log n = 0.5n \log n$ and the total share size is $0.5n \log n$. Fix a set A of size 2. With probability $1 - \frac{1}{p}$ the parties in A get two different shares and can reconstruct the secret. Notice that the correctness of the scheme is not perfect, since, with probability $1/p$, two-party subsets cannot reconstruct the secret. △

5 Multi-linear Schemes and q-Polynomial Schemes

In this section, we show that any r-linear secret sharing scheme over $\mathbb{F}_q$ is a polynomial secret sharing scheme over $\mathbb{F}_{q^r}$, with both sharing and reconstruction is by polynomials of degree at most q^{r-1}.

Towards proving this result, we study a particular family of polynomial schemes, where the sharing polynomials are the so-called q-polynomials. Next, we define this family of polynomials, and we show interesting properties of the associated schemes. Namely, we provide a connection between this family of polynomial schemes and multi-linear schemes, and we provide examples that illustrate the need for restrictions on the degree of polynomials in Theorem 3.2. For more details about matroids represented by q-polynomials, see [27,62].

Definition 5.1. *Let q be a prime power and consider the finite field $\mathbb{F}_{q^r}$ for some $r > 0$. A polynomial of the form $P(x) = \sum_{i=0}^{r-1} a_i x^{q^i}$ where $a_i \in \mathbb{F}_{q^r}$ is a q-polynomial.*

Equipped with the operations of addition and composition of polynomials in $\mathbb{F}_{q^r}[x]$ modulo $x^{q^r} - x$, the set of q-polynomials forms a $\mathbb{F}_q$-algebra that is isomorphic to the algebra of $r \times r$ matrices over $\mathbb{F}_q$ (see [70,32]). These polynomials were first studied by Ore in [70] where they showed a structure of a skew field, i.e., a noncommutative ring with a field of fractions. Later, Carlitz [32] described an isomorphism between the $\mathbb{F}_q$-linearized polynomial mappings over $\mathbb{F}_{q^r}$ and the group of $r \times r$ matrices over $\mathbb{F}_{q^r}$. However, the mentioned papers only study univariate linearized polynomials, while our sharing polynomials are polynomials over more than one variable. In Proposition B.1 we give the details of the construction given in [21] of the isomorphism between multivariate linearized polynomials and multivariate q-polynomials over $\mathbb{F}_{q^r}$, which are sum of univariate q-polynomials. Using an isomorphism from [21], we obtain the following correspondence between multi-linear secret sharing schemes and polynomial schemes with q-polynomial sharing.

Theorem 5.2. *Every r-linear secret sharing scheme over $\mathbb{F}_q$ is a polynomial secret sharing scheme over $\mathbb{F}_{q^r}$ where the sharing and reconstruction polynomials are q-polynomials of degrees at most q^{r-1}. Conversely, any (perfect) polynomial secret sharing scheme over $\mathbb{F}_{q^r}$ in which the sharing polynomials are q-polynomials is a multi-linear secret sharing scheme.*

Remark 5.3. As we observed in Remark 2.12, every ideal secret sharing scheme whose domain of secrets and shares has size q^r is a polynomial secret sharing scheme over $\mathbb{F}_{q^r}$, where the degree of each variable in the sharing polynomials is at most $q^r - 1$. In the case of ideal r-linear secret sharing schemes over $\mathbb{F}_q$, the total degrees of the sharing and reconstruction polynomials are bounded by q^{r-1}.

Polynomial secret sharing schemes with q-polynomial sharing are not perfect in general (see Example 5.6). However, the properties that characterize perfect multi-linear secret sharing schemes can be translated to the q-polynomials induced by the linear mappings. Hence, a set of q-polynomials defines a perfect secret sharing scheme over $\mathbb{F}_{q^r}$ if and only if the associated r-linear scheme is perfect. When restricting to the ideal setting we get the following characterization.

Remark 5.4. Let $P_1, \ldots, P_n$ be q-polynomials and $P_0(s, r) = s$. If the polynomial secret sharing scheme over $\mathbb{F}_{q^r}$ defined by $P_0, P_1, \ldots, P_n$ is ideal, then there exist $P'_1, \ldots, P'_n$ q-polynomials that realize the same access structure and each P'_i is a permutation polynomial in every variable of its support for $1 \leq i \leq n$ (i.e., when restricting to every variable, the polynomial mapping is a permutation of $\mathbb{F}_q$). See [11] for more details.

5.1 Examples of q-Polynomial Schemes

In Example 1.1 and Example 1.11 we have shown examples of ideal q-polynomial secret sharing schemes. In Example 5.5 we see the construction of the matrix that represents a multi-linear scheme deduced from an ideal q-polynomial scheme. However, schemes with q-polynomial sharing are not perfect, in general. In Example 5.6 we describe a q-polynomial scheme that is not perfect and it induces a non-ideal multi-linear scheme.

At the end we give an example of a scheme with q-polynomial sharing that does not realize the access structure determined by the algebraic dependence of the sharing polynomials however it is a perfect multi-linear scheme for another access structure. This example shows the importance of the restriction on the degree of the sharing and reconstruction polynomials with respect to the size of the field as given in Theorem 3.2.

Example 5.5. We consider the scheme of the Example 1.11 with sharing polynomials over $\mathbb{F}_4$ $P_0(x, y, z) = x$, $P_1(x, y, z) = y + z^2$, $P_2(x, y, z) = x + y^2 + z$. This is, the scheme with secret P_0 and shares computed by the polynomials P_1, P_2 over $\mathbb{F}_4$. This scheme determines a 2-linear scheme over $\mathbb{F}_2$ as follows. Let α be a generator of $\mathbb{F}_4$ over $\mathbb{F}_2$ satisfying $\alpha^2 + \alpha + 1 = 0$. We consider the basis $\{1, \alpha\}$ of $\mathbb{F}_4$ as a $\mathbb{F}_2$-vector space. The mappings $z \mapsto az$ and $z \mapsto z^2$, with $a \in \mathbb{F}_2$ are linear over $\mathbb{F}_2$. Indeed, let $z = a_1 + a_2\alpha \in \mathbb{F}_4$ with $a_1, a_2 \in \mathbb{F}_2$, then $a_1^2 = a_1$, $a_2^2 = a_2$ and $\alpha^2 = \alpha + 1$, so $(a_1 + a_2\alpha)^2 = a_1 + a_2 + a_2\alpha$. Therefore, the mapping $z \mapsto z^2$ can be seen as a $\mathbb{F}_2$-linear map as

$$(a_1, a_2) \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

Then, the 2-linear representation of the scheme is defined by the $\mathbb{F}_2$-linear maps with matrices:

$$M_{P_0} = \begin{pmatrix} 1\,0 \\ 0\,1 \\ 0\,0 \\ 0\,0 \\ 0\,0 \\ 0\,0 \end{pmatrix} \quad M_{P_1} = \begin{pmatrix} 0\,0 \\ 0\,0 \\ 1\,0 \\ 0\,1 \\ 1\,0 \\ 1\,1 \end{pmatrix} \quad M_{P_2} = \begin{pmatrix} 1\,0 \\ 0\,1 \\ 1\,0 \\ 1\,1 \\ 1\,0 \\ 0\,1 \end{pmatrix}.$$

We can observe that the matrix $M_{P_0}|M_{P_1}|M_{P_2}$ defines an ideal 2-linear scheme; therefore, the polynomial scheme with secret P_0 and two parties with sharing P_1 and P_2 is ideal. We can check that it is a multi-linear scheme for the port of the uniform matroid U_3^2.

Nevertheless, notice that the algebraic matroid defined by the polynomials is the uniform matroid U_3^3 since the algebraic rank of P_0, P_1, and P_2, is 3, i.e., they are an independent set of the matroid. Therefore, the polynomial scheme does not realize the access structure determined by the port of the algebraic matroid. $\triangle$

Example 5.6. Consider now the scheme over $\mathbb{F}_4$ where the secret corresponds to the polynomial $P_0(x, y) = x$ and the shares are computed by the polynomials:

$$P_0(x, y) = x, \quad P_1(x, y) = x^2 + x + y, \quad P_2(x, y) = y.$$

This scheme does not satisfy correctness since the parties P_1 and P_2 cannot fully recover the secret since they recover $x^2 + x$ that is not invertible over $\mathbb{F}_4$. In terms of the multi-linear scheme, the linear mappings defined by the polynomials are:

$$M_{P_0} = \begin{pmatrix} 1\;0 \\ 0\;1 \\ 0\;0 \\ 0\,0 \end{pmatrix} \quad M_{P_1} = \begin{pmatrix} 0\,0 \\ 1\,0 \\ 1\,0 \\ 0\,1 \end{pmatrix} \quad M_{P_2} = \begin{pmatrix} 0\,0 \\ 0\,0 \\ 1\,0 \\ 0\,1 \end{pmatrix}.$$

Note that M does not define an ideal multi-linear scheme since the matrix $M_{P_1}|M_{P_2}$ has rank 3. Therefore, neither the polynomial scheme nor the multi-linear scheme is ideal. $\triangle$

In previous examples (see Example 1.11), the access structure of a polynomial scheme may not be determined by the algebraic dependence of the sharing polynomials. This justifies the need for restrictions on the degree of polynomials in Theorem 3.2. We next see an example of an ideal q-polynomial scheme that shows the need for the bound on the degree of the sharing and reconstruction polynomials of Theorem 3.2 for an arbitrary number of parties.

Example 5.7. Let q be a prime power, n be the number of parties, and q be a prime power. We define a q-polynomial scheme over $\mathbb{F}_{q^n}$ with secret $P_0 = x_0$ and shares computed by the polynomials

$$\begin{aligned} P_1 &= x_0 + x_1 - x_2^q, \\ P_i &= x_i - x_{i+1}^q \text{ for } 1 < i < n, \text{ and} \\ P_n &= x_n - x_1^q. \end{aligned}$$

To prove that the scheme with these sharing is ideal, we can construct the correspondent multi-linear representation and observe that it gives a representation for an ideal scheme.

Now, we show that the access structure does not correspond to the one determined by the algebraic dependence of P_i on P_0. Observe that the set of all parties is authorized since there is a reconstruction function

$$Q(y_1, \ldots y_n) = y_1 + y_2^q + y_3^{q^2} + \ldots + y_{n-1}^{q^{n-2}} + y_n^{q^{n-1}}$$

satisfying that for all $x_0, \ldots, x_n \in \mathbb{F}_{q^n}$,

$$Q(P_1(x_0, \ldots, x_n), \ldots, P_n(x_0, \ldots, x_n)) = x_0$$

since $x_1 - x_1^{q^n} = 0$ for all $x_1 \in \mathbb{F}_{q^n}$. However, the set of polynomials $P_0, \ldots, P_n$ is algebraically independent, since the Jacobian matrix $J_{(x_0,\ldots,x_n)}(P_0, \ldots, P_n)$ has rank $n+1$:

$$J_{(x_0,\ldots,x_n)}(P_0, \ldots, P_n) = \begin{pmatrix} 1 & 0 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ 0 & 1 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \end{pmatrix}$$

Here we use the implication of the Jacobian criterion that is true for arbitrary characteristic, see [9].

Notice that the maximum degree of the sharing polynomials is $d_1 = q$ and the reconstruction polynomial has degree $d_2 = q^{n-1}$. As subsets with less than n parties are all unauthorized, there is only one reconstruction polynomial.

Therefore, we have described a scheme over $\mathbb{F}_{q^n}$ such that its access structure is not the port of the algebraic matroid defined by the sharing polynomials and the bound is $q^n = \max\{q^n, q \cdot q^{n-1}\}$. △

6 Secret Sharing Schemes from Algebraic Varieties

In this section, we explore a new family of secret sharing schemes whose shares are determined by points on an algebraic variety as done in the work of Matúš [69]. We show good security properties on these schemes and then, using results from Jafari and Khazaei [54], we construct statistical schemes for the access structure determined by the polynomials.

In the following section we provide the definition of a statistical scheme and the construction of [54].

6.1 Statistical Secret Sharing Schemes

In the definition of a secret sharing scheme realizing an access structure, the correctness and privacy are perfect. We can relax these requirements and ask that for every unauthorized set B and two secrets a, b, the correctness holds with high probability and that the statistical distance between $\Pi_B(a; r)$ and $\Pi_B(b; r)$ is small. To quantify high probability and small statistical distance, the scheme gets as a security parameter ℓ (in unary) as an additional input.

Definition 6.1 (Statistical SSS). *Let ℓ be a security parameter. A statistical secret sharing $\Sigma = \langle \Pi, \mu \rangle$ is a pair where, for every $\ell \in \mathbb{N}$, $\Pi(1^\ell, \cdot, \cdot)$ is a mapping from $[K_0(\ell)] \times [K_R(\ell)]$ to $[K_1(\ell)] \times \cdots \times [K_n(\ell)]$ for some functions $K_0, K_R, K_1, \ldots, K_n : \mathbb{N} \to \mathbb{N}$, and μ is a collection of distributions on $K_R(\ell)$. We say that Σ is a* statistical secret sharing scheme *for the access structure Γ (or Σ statistically realizes it) if:*

Polynomial secret length growth. *The secret length grows at most polynomially in ℓ,*

$$\log K_0(\ell) = O(\ell^c) \text{ for some } c \in \mathbb{N}.$$

Statistical Correctness. *There is some $\varepsilon(\ell) \in \ell^{-\omega(1)}$ (i.e., a negligible function in ℓ) such that for every qualified set $A \in \Gamma$, there exists a reconstruction function $\textsc{Recon}_A$ such that for every secret s and every security parameter ℓ,*

$$\mathsf{Pr}(\textsc{Recon}_A(\Pi(1^\ell, s, r)_A) = s) \geq 1 - \varepsilon(\ell),$$

where the probability is over the choice of r according to $\mu(1^\ell)$

Statistical Privacy. *There is some $\varepsilon(\ell) \in \ell^{-\omega(1)}$ (i.e., a negligible function in ℓ), such that for every unauthorized set $B \notin \Gamma$ and for every pair of secrets a, b*

$$\mathsf{SD}(\Pi(1^\ell, a, r)_B, \Pi(1^\ell, b, r)_B) \leq \varepsilon(\ell).$$

We define the optimal information ratio of statistical schemes *for an access structure Γ as the infimum of the information ratio of all sequences of statistical schemes realizing Γ.*

The following result guarantees the existence of statistical secret sharing schemes from a family of random variables. It is obtained by combining different results from [54]. The proof of Proposition 6.2 is included in Appendix E.2 for the sake of completeness. We use this result as a black-box transformation in Proposition 6.8.

Proposition 6.2 ([54]). *Let $S_0, \ldots, S_n$ be jointly distributed random variables and let Γ be an access structure on $\{1, \ldots, n\}$. If*

$$C = \min_{B \notin \Gamma} H(S_0|S_B) - \max_{A \in \Gamma} H(S_0|S_A) > 0,$$

then there exists a sequence of random variables $(T_0^\ell, \ldots, T_n^\ell)_{\ell \in \mathbb{N}}$ satisfying that $|\mathrm{supp}(T_0^\ell)| = 2^{\ell R}$, where $R = C - \Theta(\ell^{-1/4})$, with the following properties: For *every $s \in \mathrm{supp}(T_0^\ell)$ and for every subset A in Γ there exists a reconstruction function* $\textsc{Recon}_A$ *with*

$$\mathsf{Pr}(\textsc{Recon}_A(T_A^\ell) \neq T_0^\ell | T_0^\ell = s) < 2e^{-\sqrt{\ell}}. \tag{4}$$

and for every $s \in \mathrm{supp}(T_0^\ell)$ and for every subset B not in Γ,

$$\mathsf{SD}(T_B^\ell | T_0^\ell = s, T_B^\ell) \leq 6e^{-\sqrt{\ell}}. \tag{5}$$

Indeed, the schemes Σ with Π defined as the outcome of the random variables $(T_1^\ell, \ldots, T_n^\ell)_\ell$, when T_0^ℓ is taken uniformly at random, is a statistical scheme for Γ with information ratio $\max_{i \in P} H(S_i)/C$ and security parameter ℓ. The share size of i-th share is $\ell \log |S_i|$ and secret size ℓR.

In Appendix D and Appendix E.1 we give more details about this transformation, which uses wiretap channel techniques.

6.2 Schemes from Algebraic Varieties

In Theorem 4.8 we discuss the constructions of secret sharing schemes for access structures determined by polynomials by considering polynomial sharing by evaluating the polynomials. In this section, we consider the algebraic variety defined by the ideal of annihilator polynomials and give as shares a uniformly distributed point of the variety.

Using a new result in matroid theory by Matúš [69], we construct sequences of secret sharing schemes with information ratio tending to one from algebraic matroids. Afterwards, using the transformation by Jafari and Khazaei [54] (i.e., Proposition 6.2), we obtain statistical schemes with information ratio tending to one as the security parameter grows. This result does not provide any bound on the secret size. Next, we explain the ideas of this construction in more detail; this construction follows in three steps. Most of the technical results and all proofs appear in Appendix E.

Theorem 6.3 (Theorem 1.9 Restated). *Every port of an algebraic matroid over a finite field admits a statistical secret sharing scheme with information ratio tending to* 1*.*

Next, we introduce some definitions and results needed for the proof of Theorem 6.3. A *polymatroid* is a pair (E, r) where E is a finite set and $r : 2^E \to \mathbb{R}$ is a rank function satisfying $r(\emptyset) = 0$, and the conditions *2* and *3* of Definition 2.5. I.e., if $X \subseteq Y$ then $r(X) \leq r(Y)$ and for every $X, Y \subseteq E$, $r(X \cap Y) + r(X \cup Y) \leq r(X) + r(Y)$. Hence, a polymatroid generalizes matroids by not requiring that $r(X)$ is an integer bounded by $|X|$.

Theorem 6.4 ([49,48]). *Let $(S_x)_{x \in E}$ be some random variables and, for every $X \subseteq E$, define $S_X = (S_x)_{x \in X}$. Consider the mapping $h\colon 2^E \to \mathbb{R}$ defined by $h(\emptyset) = 0$ and $h(X) = H(S_X)$ if $\emptyset \neq X \subseteq E$. Then, h is the rank function of a polymatroid with ground set E.*

Polymatroids that can be defined from random variables as in Theorem 6.4 are called *entropic*. Matúš [69] proved that for every algebraic matroid $\mathcal{M} = (E, r)$, there exists a sequence of entropic polymatroids whose rank function tends to a multiple of the matroid rank r, i.e., algebraic matroids are *almost entropic*. In this construction, the polymatroids are determined by the algebraic variety defined by the annihilating polynomials of the algebraic representations of the matroid. Recall that the variety of a set of polynomials $P_1, \ldots, P_m \in \mathbb{F}[x_1, \ldots, x_n]$ is the set $(a_1, \ldots, a_n) \in \mathbb{F}^n$ such that $P_i(a_1, \ldots, a_n) = 0$ for every $1 \leq i \leq m$.

Let $\mathcal{M}$ be an algebraic matroid over a finite field $\mathbb{G}$. For a large enough extension $\mathbb{F}$ of $\mathbb{G}$, the variety defined by the annihilating polynomials of the algebraic representation of $\mathcal{M}$ is non empty, and we define the random variables $S_{\mathbb{F}}$ that take uniform distribution on the points of this variety. Define the jointly distributed random variables $S_{\mathbb{F},A}$ that take the projection of $S_{\mathbb{F}}$ over the coordinates in A, for every $A \subseteq E$. Define the polymatroid $\mathcal{S}_{\mathbb{F}} = (E, h_{\mathbb{F}})$ with rank function

$$h_{\mathbb{F}}(A) = \frac{H(S_{\mathbb{F},A})}{\log |\mathbb{F}|}.$$

The sequence of entropic polymatroids $\mathcal{S}_{\mathbb{F}}$ is obtained by considering an increasing sequence of algebraic varieties over field extensions of $\mathbb{F}$. Define

$$\sigma_{\mathbb{F}} = \frac{\max_{i \in P} h_{\mathbb{F}}(P_i)}{h_{\mathbb{F}}(P_0)},$$

where the maximum is taken for $i \in E \setminus \{0\}$.

The proof of Lemma 6.5, which is from [69], can be found in Appendix C.

Lemma 6.5. *Let $\mathbb{F}_q$ be a finite field of size q and $\mathcal{S}_{\mathbb{F}_q} = (E, h_{\mathbb{F}_q})$ be the polymatroid defined above. Then, for every $K \subseteq E$,*

$$|h_{\mathbb{F}_q}(K) - r(K)| \leq \frac{\mathfrak{n}}{\ln q}$$

for some constant $\mathfrak{n}$ that depends on the variety determined by the representation of $\mathcal{M}$, but it is independent of q.

To simplify the notation, given a polymatroid $\mathcal{S}_{\mathbb{F}} = (E, h_{\mathbb{F}})$ as defined above, we denote $h_{\mathbb{F}_q}(0 : C) = h_{\mathbb{F}_q}(0) + h_{\mathbb{F}_q}(C) - h_{\mathbb{F}_q}(C \cup \{0\})$ for any $C \subseteq E \setminus \{0\}$.

Lemma 6.6. *Let $\mathcal{M}$ be an algebraic matroid over some finite field $\mathbb{G}$ and Γ be the access structure determined by the matroid $\mathcal{M}$. The polymatroid $\mathcal{S} = (E, h_{\mathbb{F}_q}$ defined above, where $|\mathbb{F}_q| = q$, satisfies the following:*

1. *If $A \in \Gamma$, then $h_{\mathbb{F}_q}(0 : A) \geq 1 - \frac{3\mathfrak{n}}{\ln q}$.*
2. *If $B \notin \Gamma$, then $h_{\mathbb{F}_q}(0 : B) \leq \frac{2\mathfrak{n}}{\ln q}$.*

Proof. For an authorized subset $A \in \Gamma$, i.e., a set such that $r(A \cup \{0\}) = r(A)$,

$$\begin{aligned} h_{\mathbb{F}_q}(0 : A) &= h_{\mathbb{F}_q}(0) + h_{\mathbb{F}_q}(A) - h_{\mathbb{F}_q}(A \cup \{0\}) \\ &\geq 1 - \frac{\mathfrak{n}}{\ln q} + r(A) - \frac{\mathfrak{n}}{\ln q} - \left(r(A \cup \{0\}) + \frac{\mathfrak{n}}{\ln q} \right) = 1 - \frac{3\mathfrak{n}}{\ln q}. \end{aligned}$$

Let B be a non-authorized subset, this is $r(B \cup \{0\}) = 1 + r(B)$.

$$\begin{aligned} h_{\mathbb{F}_q}(0 : B) &= h_{\mathbb{F}_q}(0) + h_{\mathbb{F}_q}(B) - h_{\mathbb{F}_q}(B \cup \{0\}) \\ &\leq 1 + r(B) - \left(r(B \cup \{0\}) - \frac{\mathfrak{n}}{\ln q} \right) = \frac{2\mathfrak{n}}{\ln q} \end{aligned}$$

□

Combining this technical lemma with the construction detailed in Appendix C, we deduce the following result.

Proposition 6.7. *Let $\mathcal{M}$ be an algebraic matroid over a finite field $\mathbb{G}$, and let Γ be a port of $\mathcal{M}$. For a sufficiently large extension $\mathbb{F}$ of $\mathbb{G}$, the random variables $S_{\mathbb{F},A}$ satisfy that*

$$C = \min_{B \notin \Gamma} H(S_{\mathbb{F},0} | S_{\mathbb{F},B}) - \max_{A \in \Gamma} H(S_{\mathbb{F},0} | S_{\mathbb{F},A}) > 0.$$

Proof. Since $1 \geq h_{\mathbb{F}}(0) \geq 1 - \frac{\mathfrak{n}}{\ln q}$, by Lemma 6.6, the polymatroid $\mathcal{S}_{\mathbb{F}} = (E, h_{\mathbb{F}})$ satisfies that

$$\begin{aligned} \max_{A \in \Gamma} h_{\mathbb{F}}(A \cup \{0\}) - h_{\mathbb{F}}(A) &= \max_{A \in \Gamma} h_{\mathbb{F}}(0) - h_{\mathbb{F}}(0 : A) \leq \frac{\mathfrak{n}}{\ln q}, \\ \min_{B \notin \Gamma} h_{\mathbb{F}}(B \cup \{0\}) - h_{\mathbb{F}}(B) &= \min_{B \notin \Gamma} h_{\mathbb{F}}(0) - h_{\mathbb{F}}(0 : B) \geq 1 - \frac{2\mathfrak{n}}{\ln q}. \end{aligned}$$

Therefore,

$$\begin{aligned} C =& \log |\mathbb{F}| \left(\min_{B \notin \Gamma} (h_{\mathbb{F}}(B \cup \{0\}) - h_{\mathbb{F}}(B)) - \max_{A \in \Gamma} (h_{\mathbb{F}}(A \cup \{0\}) - h_{\mathbb{F}}(A)) \right) \\ \geq& 1 - O\left(\frac{1}{\ln q} \right). \end{aligned}$$

So if the field $\mathbb{F}$ is sufficiently large, $C > 0$. □

These random variables define a partial secret sharing scheme when $\mathbb{F}$ is large enough. Next, we use the black-box constructions of [54] (described in Proposition 6.2) transforming partial secret sharing schemes to statistical secret sharing schemes.

Proposition 6.8. *Let $\mathcal{M}$ be an algebraic matroid over a finite field $\mathbb{G}$, and let Γ be a port of $\mathcal{M}$. There is a statistical secret sharing scheme realizing Γ whose information ratio tends to $\sigma_{\mathbb{F}}$ by increasing the security parameter (where the secret domain is $\mathbb{F}$ for a sufficiently large extension $\mathbb{F}$ of $\mathbb{G}$).*

The proof of this proposition is by combining Proposition 6.2 and Proposition 6.7. Observe that $\sigma_{\mathbb{F}}$ tends to 1 when $|\mathbb{F}| \to \infty$ because

$$1 - O\left(\frac{1}{\ln q}\right) \leq \sigma_{\mathbb{F}} \leq 1 + O\left(\frac{1}{\ln q}\right).$$

Next, following the ideas in [54], we construct a statistical secret sharing scheme with information ratio tending to 1. To do that, we consider a field $\mathbb{F}$ for every security parameter ℓ, such that $\sigma_{\mathbb{F}}$ tends to 1 when ℓ increases. This completes the proof of Theorem 6.3. This construction is detailed in Appendix E.3.

In Theorem 1.8, we discuss the case where each party may have more than one polynomial. We consider the statistical scheme of Theorem 6.3 in an extended set of parties $\{1, \ldots, k\}$ where each party has one polynomial. We use this scheme on the set of n parties by giving to each party i, the k_i shares of the scheme. Then, it has total information ratio tending to k when the security parameter increases.

6.3 Statistical Schemes from Polynomials

Under the hypothesis of Theorem 4.8, we can also construct schemes with statistical security using the black-box transformation of Proposition 6.2 above.

Theorem 6.9. *Let t be the size of the largest forbidden set in Γ, $t > \sum_{i \in A} k_i$, and $\log p > (t^2 + 3t + 2) \log d$, then there exists a statistical scheme Σ realizing Γ with total share size $k\ell \log p$ and secret size $\ell(C - \Theta(\ell^{-1/4}))$, where $C = (1 - td^t/p) \log p - 3t^2 \log d$ and ℓ is the security parameter.*

First, we prove this result when every party has only one polynomial. In this case, we denote the polynomials simply as $P_0, \ldots, P_n$. The result for the general case is obtained by combining Theorem 4.8 and Proposition 6.2 as follows.

Consider the jointly distributed random variables $S_A = \{P_i(U_m)\}_{i \in A}$, where U_m is the uniform distribution over $\mathbb{F}_p^{m+1}$. By Theorem 4.8, they have positive secret capacity C. By Proposition 6.2, there exists a statistical scheme Σ realizing Γ with shares T_i^ℓ and secret space T_0^ℓ of size $2^{\ell R}$, with $R = C - \Theta(\ell^{-1/4})$ and security parameter ℓ.

Now we consider the case, where each party may have more than one polynomial and there are k polynomials in total. Notice that in this case, we can consider a scheme with an extended set of parties $\{1, \ldots, k\}$, where each party has one polynomial. Now, the subsets that are forbidden in Γ correspond to subsets of size less than t in this new setting. Applying the results we got above, we obtain a statistical secret sharing scheme with security parameter ℓ realizing Γ with total share size $k\ell \log p$ and same secret size.

In the case that Γ is a t-out-of-n threshold access structure, then the result can be slightly improved. It is enough to require $\log p > 6t \log d$ to get statistical schemes with secret size $\ell(C - \Theta(\ell^{-1/4}))$, where $C = (1 - td^t/p) \log p - 5t \log d$.

Our construction of statistical secret sharing schemes from polynomials in Theorem 6.9 has a restriction on the degree of the polynomials and the size of the field. Here we show some examples of schemes which sharing polynomials do not satisfy this restriction and they lead to non valid schemes for their access structures.

Example 6.10. Let $p = 2q + 1$ be a safe prime, i.e., q is prime. Then, the polynomial scheme in $\mathbb{F}_p$ with $P_1(s, r) = r$, $P_2(s, r) = s^q + r$ and $P_3(s, r) = r^q + s$ as polynomial sharings does not satisfy the hypothesis of Theorem 1.7 since the degree of the polynomials exceeds the bound needed, $\log p > 12 \log q$.

The access structure determined by these polynomials is the 2-out-of-n access structure. However, the third party has much more information about the secret for any $s, r \in \mathbb{F}_p$ since $r^q = \pm 1$. Moreover, the authorized set defined by containing the first and second party has almost no information about the secret since they can recover s^q which takes only ± 1 values. Therefore, there is a forbidden set that has much more information about the secret than an authorized set. △

6.4 Duality of Statistical Secret Sharing Schemes

A classical and open question about secret sharing schemes is if duality preserves the optimal information ratio of an access structure or the optimal share size, even for the ideal case. Recall that the dual of an access structure Γ is $\Gamma^* = \{A \subseteq E : E \setminus A \notin \Gamma\}$. When the schemes are linear or multi-linear, the information ratio is preserved by duality [71]. Our work about statistical secret schemes for access structures determined by algebraic matroids together with the fact that there exist almost entropic matroids whose dual does not satisfy this property [36,56] prove Theorem 1.10.

Proof of Theorem 1.10. Kaced [56] proved that the class of almost entropic matroids is not closed by duality, and an explicit counterexample was presented by Csirmaz [36].

If a matroid $\mathcal{M}$ is almost entropic, there exists a sequence of entropic polymatroids that tend to the matroid. If Γ is a port of $\mathcal{M}$ at 0, this sequence of polymatroids satisfies that the entropy of 0 conditioned to an authorized set is very low and the entropy of 0 conditioned to a forbidden set is almost the entropy of 0. Using the transformation in Proposition 6.2, we can transform these polymatroids to a family of statistical secret sharing schemes whose information ratio tends to 1.

Conversely, if we have statistical secret sharing schemes with information ratio tending to one, they define a sequence of polymatroids that tends to an entropic matroid. Then, the access structure of the scheme is a port of a matroid that is almost entropic.

Therefore, access structures that are ports of $\mathcal{M}$ have optimal information ratio 1, while ports of its dual have optimal information ratio strictly greater than 1. □

Since it is not true that all the access structures preserve optimal information ratio for statistical schemes by duality, a natural question is to characterize the family of matroids for which this property holds. By the results we obtained, for ports of algebraic matroids, the optimal information ratio for statistical schemes is 1. However, it is not known if algebraic representation is closed under duality, in general.

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A Wooley's Theorem

We dedicate this section to the proof of Theorem 4.4, which is a corollary of Wooley's theorem.

Proof of Theorem 4.4. To prove the result for $m > k$, we define m polynomials by fixing the values of the first $m - k$ variables. As a first step, fix any $a_{k+1}, \ldots, a_m \in \mathbb{F}_p$. Define the polynomials $P'_1, \ldots, P'_m$, where $P'_i(x_1, \ldots, x_m) = P_i(x_1, \ldots, x_m) - a_i$ for $1 \leq i \leq k$ and $P'_i(x_1, \ldots, x_m) = x_i - a_i$ for $k+1 \leq i \leq m$. Define $J = J_x(P_1, \ldots, P_k)$ and $J' = J_x(P'_1, \ldots, P'_m)$; thus,

$$J' = \begin{pmatrix} & J \\ 0 & \mathrm{Id}_{m-k} \end{pmatrix}.$$

The Jacobian $J'(c)$ has full rank if and only if the first k columns of $J(c)$ are independent. We apply Wooley's theorem (Theorem 4.3) for $P'_1, \ldots, P'_m$ and conclude that $N \leq d^k$.

Since there are p^{m-k} possible values for $a_{k+1}, \ldots, a_m \in \mathbb{F}_p$, then $N_{a_1,\ldots,a_k} \leq d^k \cdot p^{d-k}$. $\square$

B q-Polynomial Mappings

We dedicate this section to the following result, which establishes a correspondence between q-polynomial and linear mappings.

Proposition B.1. *Let q be a prime power and $\mathbb{F}_{q^r}$ a finite field. The algebra of q-polynomials on m variables over $\mathbb{F}_{q^r}$ is isomorphic with the algebra of $\mathbb{F}_q$-linear maps from $\mathbb{F}_q^{rm}$ to $\mathbb{F}_q^r$.*

We prove Proposition B.1 by extending the isomorphism of [32] to the case of q-polynomials on m variables. For that, we first prove Claim B.2 and Claim B.3. The result and the proof are analogous to [21, Proposition 3.2]. We start by showing an equivalence between multivariable q-polynomials and $\mathbb{F}_q$-linear maps. Then, we see that the structure of the ring is preserved under this equivalence.

First, notice that the field $\mathbb{F}_{q^r}$ can be seen as a vector space over $\mathbb{F}_q$. Let α be a primitive element of $\mathbb{F}_{q^r}$, then $\alpha^0, \ldots, \alpha^{r-1}$ constitutes a basis of $\mathbb{F}_q^r$ over $\mathbb{F}_q$. The q-polynomials in one variable are $\mathbb{F}_q$-linear functions from $\mathbb{F}_{q^r}$ to $\mathbb{F}_{q^r}$. Then, when using the transformation of $\mathbb{F}_{q^r}$ to $\mathbb{F}_q^r$, every q-polynomial P is a $\mathbb{F}_q$-linear map from $\mathbb{F}_q^r$ to $\mathbb{F}_q^r$ with an associated $r \times r$ matrix M_P over $\mathbb{F}_q$. Indeed, the isomorphism in [32] is based on this correspondence, and this implies that every $\mathbb{F}_q$-linear map from $\mathbb{F}_q^r$ to $\mathbb{F}_q^r$ can be written as a q-polynomial over $\mathbb{F}_{q^r}$.

Claim B.2. *A q-polynomial over $\mathbb{F}_{q^r}$ defines a $\mathbb{F}_q$-linear map over the vector space $\mathbb{F}_q^r$.*

Proof. Observe first that q-polynomial mappings are $\mathbb{F}_q$-linear. That is, a q-polynomial P satisfies that if $a \in \mathbb{F}_q$ and $\alpha \in \mathbb{F}_{q^r}$, then $P(a\alpha) = aP(\alpha)$ since $a^{q^k} = a$ for every $k \geq 0$. Similarly, if $\beta \in \mathbb{F}_q$ then $P(\alpha+\beta) = P(\alpha) + P(\beta)$ since $\mathbb{F}_{q^r}$ has characteristic q and $(\alpha+\beta)^{q^k} = \alpha^{q^k} + \beta^{q^k}$.

The elements of $\mathbb{F}_{q^r}$ can be described as the $\mathbb{F}_q$-vector space of dimension r generated by the powers of α, where α is a primitive element of $\mathbb{F}_{q^r}$, i.e., a generator of the multiplicative group. Then, any q-polynomial $P \in \mathbb{F}_q[X]$ satisfies that

$$P\left(\sum_{i=0}^{r} a_i \alpha^i\right) = \sum_{i=0}^{r} a_i P(\alpha^i),$$

so, it is a $\mathbb{F}_q$-linear map and it is uniquely determined by the action of P over α^i, $1 \leq i \leq r-1$. A multivariate q-polynomial consists of sums of univariate q-polynomials; thus, it is linear in every variable and it is a $\mathbb{F}_q$-linear map over $\mathbb{F}_{q^r}$. $\square$

Next, we see that all linear maps are indeed q-polynomials. Berson observed in [21] that every multivariate $\mathbb{F}_q$-linear polynomial over $\mathbb{F}_{q^r}$ is linear in every coordinate. With this observation, we can extend the previous isomorphism to the algebra of q-polynomials in m variables. For every q-polynomial on m variables we obtain a $\mathbb{F}_q$-linear map from $\mathbb{F}_q^{rm}$ to $\mathbb{F}_q^r$, i.e., a $rm \times r$ matrix over $\mathbb{F}_q$. Since the isomorphism is maintained, for every $\mathbb{F}_q$-linear map from $\mathbb{F}_q^{rm}$ to $\mathbb{F}_q^r$ there is a corresponding q-polynomial over $\mathbb{F}_{q^r}$ in m variables.

Claim B.3. *All $\mathbb{F}_q$-linear maps from $(\mathbb{F}_q^r)^m$ to $\mathbb{F}_q^r$ are q-polynomials on m variables over $\mathbb{F}_{q^r}$.*

Proof. We first consider the case $m = 1$. Let $P \in \mathbb{F}_q^r \to \mathbb{F}_q^r$ be an $\mathbb{F}_q$-linear map. Identifying every element of $\mathbb{F}_q^r$ with an element of the finite field $\mathbb{F}_{q^r}$, it can be written as a polynomial in $\mathbb{F}_{q^r}[X]$. By linearity we have that

1. $P(X+Y) = P(X) + P(Y)$ and
2. $P(aX) = aP(X)$ for all $a \in \mathbb{F}_q$.

Let X^s be a monomial appearing in P. By linearity, if we compare terms of equal degree in the previous expressions we get

1. $(X+Y)^s = X^s + Y^s$ and
2. $(aX)^s = aX^s$ for all $a \in \mathbb{F}_q$.

Let p be the unique prime number such that q is a power of p. If s is not a power of p, say $s = dp^e$ with $d > 1$ and $e \geq 0$, where p^e is the highest power of p that divides s. Notice that $d \not\equiv 0 \mod p$. Then $(X+Y)^s = (X+Y)^{dp^e} = (X^{p^e} + Y^{p^e})^d$ contains the nonzero term $dX^{(d-1)p^e}Y^{p^e}$ which contradicts the fact that $(X+Y)^s = X^s + Y^s$. Next, we show that s is indeed a power of q and therefore P is a q-polynomial.

Let e and ℓ such that $s = p^e$ and $q = p^\ell$. Suppose that $e > 0$. We know that $a^s = a$ by linearity. Since a is a generator, $s \geq q$. Hence we can write $e = e_0\ell + e_1$

with $e_1 < \ell$ and $a^s = (a^{q^{e_0}})^{p^{e_1}} = a^{p^{e_1}}$. Since q is the smallest integer bigger than 1 that satisfies $a^q = a$ and $p^{e_1} < q$, we have that $e_0 = 1$ and s is a power of q.

Now we consider the case $m > 1$. Recall that multivariable q-polynomials are just the sum of univariate q-polynomials. Let F be a linear map from $(\mathbb{F}_q^r)^m$ to $\mathbb{F}_q^r$. By linearity, the sum of every q-polynomial obtained when restricting to every coordinate F_i, $1 \le i \le m$, is the whole map F. Therefore F is a q-polynomial on m variables. $\square$

Proof of Proposition B.1. The set of q-polynomials is a non-commutative ring with the product defined as the composition of polynomials. We denote it as $\mathbb{F}_{q^r}^{(q)}[X]$. Indeed, if P, Q are q-polynomials, then $P * Q(z) := P(Q(z))$ is also a q-polynomial. Consider the morphism of rings $\mathbb{F}_{q^r}^{(q)}[X] \to \mathbb{F}_q^{r\times r}$ consisting of the transformation of a q-polynomial P to the matrix M_P determined by the linear mapping induced in P in Claim B.2. This is well defined since if P, Q are q-polynomials, $M_{P+Q} = M_P + M_Q$, $M_{P*Q} = M_P \cdot M_Q$ and the identity q-polynomial goes to the identity $r \times r$ matrix Id_r.

The kernel of this morphism is the set of q-polynomials that is 0 when evaluated at every $x \in \mathbb{F}_{q^r}$. This is, the ideal generated by $X^{q^r} - X$. From B.3 we note that it is a surjective morphism. Therefore,

$$\mathbb{F}_{q^r}^{(q)}[X]/(X^{q^r} - X) \simeq \mathbb{F}_q^{r\times r}.$$

Then, the set of q-polynomials under the operation of composition modulo $X^{q^r} - X$ constitutes a group isomorphic to the $r \times r$ matrices over $\mathbb{F}_q$.

This isomorphism can be extended to q-polynomials on m variables and $mr \times r$ matrices:

$$\mathbb{F}_{q^r}^{(q)}[X_1, \dots, X_m]/(X_1^{q^r} - X_1, \dots, X_m^{q^r} - X_m) \simeq \mathbb{F}_q^{mr\times r}$$

deducing $\mathbb{F}_q$-linear maps from $\mathbb{F}_q^{rm}$ to $\mathbb{F}_q^r$. $\square$

Now we revisit the proof of Theorem 3.2 but in the case of having q-polynomial sharing. We improve the bound on the degree of the polynomials.

Corollary B.4. *Let Σ be an ideal polynomial secret sharing scheme realizing an n-party access structure Γ. Suppose that the domain of secrets in Σ is $\mathbb{F}_{q^r}$ for some prime power q and the sharing and reconstruction are q-polynomials over $\mathbb{F}_{q^r}$ of degree at most d_1 and d_2 respectively. If $q^r > \max\{d_1^n, d_1 d_2\}$, then Γ is a port of the algebraic matroid defined by the sharing polynomials of Σ.*

Proof. The difference in this result is the item 2 of the proof, i.e., the case where $A \cup \{0\}$ is a circuit in the matroid $\mathcal{M}$. Let $P_0 = s$ and $\{P_i\}_{i\in A}$ be the polynomials of the sharing. By Theorem 2.14, there exists an annihilating polynomial $F \in \mathbb{F}[y_0, y_1, \dots, y_{|A|}]$ with $\deg(F) \le d_1^{|A|}$, i.e.

$$F(P_0, \{P_i\}_{i\in A}) = 0.$$

Moreover, this annihilating polynomial is again a q-polynomial, therefore a linear polynomial on every variable,

$$F = F_{P_0} + \sum_{i \in A} F_{P_i}.$$

Now, consider the polynomial G with variables $s, r_1, \ldots, r_m$ defined as

$$\begin{aligned} G(s, r_1, \ldots, r_m) &= F(0, (P_i(s, r_1, \ldots, r_m))_{i \in A}) = \\ &= \sum_{i \in A} F_{P_i}(s, r_1, \ldots, r_m) = -F_{P_0}(s, r_1, \ldots, r_m). \end{aligned}$$

The degree of G is bounded by the product of degrees

$$\deg(G) \leq \deg(F) \cdot \deg P_0 \leq (d_1)^{|A|} \leq d_1^n < q^r.$$

The rest of the proof follows as the original proof of the Theorem 3.2. □

C Algebraic Matroids are Almost Entropic

In this section, we first provide details of the proof of the following result by Matúš [69]. Then, using a construction in that proof, we are able to prove some technical results given in Section 6.

Theorem C.1 ([69]). *Algebraic matroids are almost entropic.*

The proof of Theorem C.1 is constructive. Given an algebraic matroid $\mathcal{M} = (E, r)$, and the access structure given by the matroid port Γ, a sequence of random variables $\{S_q\}_q$ is constructed such that, for q big enough, the polymatroids defined by the distributions of those random variables partially realize the access structure Γ.

First, consider the algebraic representation of $\mathcal{M}$. That is, $n = |E|$, $\mathbb{G}$ a finite field, and $\mathbb{H}$ a transcendental extension of $\mathbb{G}$ that contains the algebraic elements representing $\mathcal{M}$, $(e_i)_{i \in E} \subseteq \mathbb{H}$. Then,

$$r(A) = \deg_{tr/\mathbb{G}} \mathbb{G}((e_i)_{i \in A})$$

for every $A \subseteq E$. Now, consider the polynomials with coefficients in $\mathbb{G}$ that vanish after substituting e_i, where $i \in E$, and the ideal generated by these polynomials, $I_{\mathbb{G}}$. This is a prime ideal since if a product of two polynomials belongs to $I_{\mathbb{G}}$, then one of them vanishes after substituting e_i, $i \in E$. Consider the affine algebraic variety $V_{\mathbb{G}}$ in $\mathbb{G}^n$ defined by the ideal $I_{\mathbb{G}}$. If we replace $\mathbb{G}$ by $\overline{\mathbb{G}}$ we still get a prime ideal $I_{\overline{\mathbb{G}}}$ and the variety $V_{\overline{\mathbb{G}}}$ satisfies that $I(V_{\overline{\mathbb{G}}}) = I_{\overline{\mathbb{G}}}$ and therefore

$$\mathbb{G}[x_1, \ldots, x_n]/I_{\overline{\mathbb{G}}} \simeq \mathbb{G}[e_1, \ldots, e_n].$$

The dimension of the variety $V_{\overline{\mathbb{G}}}$ is the transcendence degree of its function field, which is the field of fractions of $\mathbb{G}[e_1, \ldots, e_n]/I_{\overline{\mathbb{G}}}$. Taking transcendence degree over $\mathbb{G}$, we get

$$\dim V_{\overline{\mathbb{G}}} = \deg_{tr/\mathbb{G}} \mathrm{Frac}(\mathbb{G}[e_1, \ldots, e_n]/I_{\overline{\mathbb{G}}}) = \deg_{tr/\mathbb{G}} \mathbb{G}((e_i)_{i \in Q}) = r(\mathcal{M}).$$

Let $\mathbb{G}_{V_{\overline{\mathbb{G}}}}$ be the smallest subfield of $\overline{\mathbb{G}}$ containing all the coefficients of the polynomials that define $V_{\overline{\mathbb{G}}}$. Then, if $\mathbb{F}$ is a finite field extending $\mathbb{G}_{V_{\overline{\mathbb{G}}}}$, the Lang-Weil bound [61] estimates the number of points of the variety $V_{\overline{\mathbb{G}}}$ with coordinates in the field $\mathbb{F}$, debited $V_{\mathbb{F}}$. It is approximately $|\mathbb{F}|^{\dim V}$, i.e., $|\mathbb{F}|^{r(\mathcal{M})}$. That is, there exists a constant $\kappa > 0$ not depending on $\mathbb{F}$ such that

$$\left| \frac{|V_{\mathbb{F}}|}{|\mathbb{F}|^{r(\mathcal{M})}} - 1 \right| \leq \frac{\kappa}{\sqrt{|\mathbb{F}|}}. \tag{6}$$

Since $V_{\overline{\mathbb{G}}}$ is nonempty, the extension of $\mathbb{G}$ that contains the coordinates of a point in $V_{\overline{\mathbb{G}}}$ is non empty. Then, for a large enough $\mathbb{F}$, the inequality implies that $V_{\mathbb{F}}$ is not empty.

From now on, assume $V_{\mathbb{F}}$ is nonempty. Define a random variable S associated to $V_{\mathbb{F}}$ taking a uniform distribution on the coordinates of the points in $V_{\mathbb{F}}$ and let $P_{\mathbb{F}}$ be the probability measure of S. Then, for every $x_i, y_i \in \mathbb{F}$, $1 \leq i \leq n$, if $(x_1, \ldots x_n), (y_1, \ldots, y_n) \in V_{\mathbb{F}}$, then

$$P_{\mathbb{F}}(S = (x_1, \ldots, x_n)) = P_{\mathbb{F}}(S = (y_1, \ldots, y_n)).$$

Else if $(x_1, \ldots, x_n) \notin V_{\mathbb{F}}$, then $P_{\mathbb{F}}(S = (x_1, \ldots, x_n)) = 0$. For every $I \subseteq \{1, \ldots, n\}$, let $\pi_I : \mathbb{F}^n \to \mathbb{F}^I$ be the coordinate projection. The marginal probability measure on $\mathbb{F}^I$ is denoted by $P_{\mathbb{F}}^I$, and the projections of S are denoted by $S^I = \pi^I(S)$ and are random variables having joint distribution $P_{\mathbb{F}}$. This is, if $x \in \mathbb{F}^I$,

$$P_{\mathbb{F}}^I(S^I = x) = P_{\mathbb{F}}(S \in \pi_I^{-1}(x))$$

Let $\pi_{I,V_{\mathbb{F}}}^{-1}(x_I)$ be the fiber consisting of the elements in $V_{\mathbb{F}}$ that project to x_I. Then, if $x \in \pi_I(V_{\mathbb{F}})$,

$$P_{\mathbb{F}}^I(S^I = x) = \frac{|\pi_{I,V_{\mathbb{F}}}^{-1}(x_I)|}{|V_{\mathbb{F}}|}.$$

And if $x \notin \pi_I(V_{\mathbb{F}})$, it holds $P_{\mathbb{F}}^I(S^I = x) = 0$. The entropy of the marginal $P_{\mathbb{F}}^I$ can be written as

$$H(P_{\mathbb{F}}^A) = - \sum_{y_A \in \pi_A(V_{\mathbb{F}})} \frac{|\pi_{A,V_{\mathbb{F}}}^{-1}(y_A)|}{|V_{\mathbb{F}}|} \ln \frac{|\pi_{A,V_{\mathbb{F}}}^{-1}(y_A)|}{|V_{\mathbb{F}}|}.$$

Consider the polymatroid $\mathcal{M}_{\mathbb{F}} = (Q, h_{\mathbb{F}})$ with

$$h_{\mathbb{F}}(A) = H(S_A)/\log(|\mathbb{F}|) \tag{7}$$

for any $A \subseteq Q$. With the following result, the polymatroid $\mathcal{M}_{\mathbb{F}}$ can be arbitrarily close to $\mathcal{M}$ when the cardinality of $\mathbb{F}$ is sufficiently large.

Lemma C.2 (Lemma 6.5 Restated). *Let $\mathcal{M} = (E, r)$ be an algebraic matroid. Let $(E, h_{\mathbb{F}})$ be the polymatroid defined above for a finite field $\mathbb{F}$ of size q. Then, for every $K \subseteq E$,*

$$|h_{\mathbb{F}}(K) - r(K)| \leq \frac{\mathfrak{n}}{\ln q}$$

for some constant $\mathfrak{n}$ *that depends on the variety determined by the representation of* $\mathcal{M}$*, but it is independent of* q*.*

Let $q = |\mathbb{F}|$, then consider the polymatroid $\mathcal{M}_\mathbb{F} = (E, h)$ where

$$h(A) = \frac{H(P_\mathbb{F}^A)}{\ln q}. \tag{8}$$

for any $A \subseteq E$.

Lemma C.2 claims that polymatroids $\mathcal{M}_\mathbb{F}$ have good convergence properties and the rank function h tends to r when increasing q.

Proof of Lemma C.2. First, we will show that for a circuit $C \subseteq E$ and a $j \in C$,

$$h(C) - h(C \setminus j) \leq \frac{\mathfrak{j}_V}{\ln q}$$

for some constant $\mathfrak{j}_V$ not depending on the field $\mathbb{F}$. Then, we will consider an arbitrary set $K \subseteq E$ and see that

$$|h(K) - r(K)| \leq \frac{\mathfrak{n}_V}{\ln q}.$$

Let C be a circuit, the elements e_i, $i \in C$ are dependent while the e_i, $i \in C \setminus j$ are independent for all $j \in C$. Therefore, there exists a polynomial P_C with coordinates in C and irreducible, since each subset of C is independent. For a $j \in C$, this polynomial can be seen as a polynomial in the indeterminate x_j with degree $d_{j,C}$ whose coefficients are polynomials in $J = C \setminus j$.

$$H(P_\mathbb{F}^C) = H(P_\mathbb{F}^J, P_\mathbb{F}^j) = H(P_\mathbb{F}^J) + H(P_\mathbb{F}^j | P_\mathbb{F}^J)$$
$$H(P_\mathbb{F}^j | P_\mathbb{F}^J) = -\sum_{y_C \in \pi_C(V_\mathbb{F})} P_\mathbb{F}^C(S^C = y_C) \ln \frac{P_\mathbb{F}^C(S^C = y_C)}{P_\mathbb{F}^J(S^J = \pi_J^C(y_C))}$$

where $\pi_J^C : \mathbb{F}^C \to \mathbb{F}^J$ is the coordinate projector and S^C is the random variable associated with the probability distribution $P_\mathbb{F}^C$.

$$\begin{aligned} H(P_\mathbb{F}^j | P_\mathbb{F}^J) &= -\sum_{y_C \in \pi_C(V_\mathbb{F})} \frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_C)|}{|V_\mathbb{F}|} \ln \frac{\frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_C)|}{|V_\mathbb{F}|}}{\frac{|\pi_{J,V_\mathbb{F}}^{-1}(\pi_J^C(y_C))|}{|V_\mathbb{F}|}} \\ &= -\sum_{y_C \in \pi_C(V_\mathbb{F})} \frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_C)|}{|V_\mathbb{F}|} \ln \frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_C)|}{|\pi_{J,V_\mathbb{F}}^{-1}(\pi_J^C(y_C))|} \\ &= -\sum_{y_J \in \pi_J(V_\mathbb{F})} \frac{|\pi_{J,V_\mathbb{F}}^{-1}(y_J)|}{|V_\mathbb{F}|} \left[\sum_{y_j \in \mathbb{F}} \frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_j, y_J)|}{|\pi_{J,V_\mathbb{F}}^{-1}(y_J)|} \ln \frac{|\pi_{C,V_\mathbb{F}}^{-1}(y_j, y_J)|}{|\pi_{J,V_\mathbb{F}}^{-1}(y_J)|} \right] \end{aligned} \tag{9}$$

For some $y_J \in \pi_J(V_\mathbb{F})$, let $\nu_{j,C}$ be the number of $y_j \in \mathbb{F}$ such that $\pi_{J,V_\mathbb{F}}^C(y_j, y_J) = y_J$. This is the number of nonzero summands in the brackets of (9) and is at most the roots of the polynomial $P_C(y_J)(x_j)$. In the general case, the substitution of y_J to P_C results in a nonzero polynomial in x_j of degree $d_{j,C}$, then $1 \leq \nu_{j,C} \leq d_{j,C}$. Otherwise, when $P_C(y_J)(x_j) = 0$, there are at most q possible values of y_j, then $\nu_{j,C} \leq q$.

For each circuit C and $j \in C$, consider the subvariety $W_{j,C}$ of V of zeros of the coefficients of $P_C(x_j)$, which are polynomials with variables in $J = C \setminus j$. The $y_J \in \pi_J(V_{\mathbb{F}})$ such that $P_C(y_J)(x_j) = 0$ are the ones in $\pi_{J,V_{\mathbb{F}}}(W_{j,C} \cap \mathbb{F}^n)$. It follows that the sums in the brackets of (9) are dominated by

$$\sum_{y_J \in \pi_{J,V_{\mathbb{F}}}(W_{j,C} \cap \mathbb{F}^n)} \frac{|\pi^{-1}_{J,V_{\mathbb{F}}}(y_J)|}{|V_{\mathbb{F}}|} \ln q + \sum_{y_J \in \pi_{J,V_{\mathbb{F}}}((V \setminus W_{j,C}) \cap \mathbb{F}^n)} \frac{|\pi^{-1}_{J,V_{\mathbb{F}}}(y_J)|}{|V_{\mathbb{F}}|} \ln d_{j,C}$$

which is at most

$$\frac{|W_{j,C} \cap \mathbb{F}^n|}{|V_{\mathbb{F}}|} \ln q + \ln d_{j,C}.$$

Using again the Lang-Weil bound applied to the subvariety $W_{j,C}$ and V, we get

$$\frac{|W_{j,C} \cap \mathbb{F}^n|}{|V_{\mathbb{F}}|} \leq \frac{q^{\dim W_{j,C}}}{q^{r(\mathcal{M})}} \frac{c_{W_{j,C}} + \frac{\kappa_{W_{j,C}}}{\sqrt{q}}}{1 - \frac{\kappa_V}{\sqrt{q}}} \leq \frac{\mathfrak{k}_V}{q}$$

where the constant $\mathfrak{k}_V$ does not depend on the field $\mathbb{F}$. Also, take $\mathfrak{d}_V$ bigger than all $\ln d_{j,C}$ and from (9) we get

$$H(X^j | X^J) \leq \mathfrak{k}_V \frac{\ln q}{q} + \mathfrak{d}_V$$

so the non-generic part of the sum is smaller than $\mathfrak{k}_V$ and taking $\mathfrak{j}_V = \mathfrak{k}_V + \mathfrak{d}_V$ we get

$$H(P^C_{\mathbb{F}}) \leq H(P^J_{\mathbb{F}}) + \mathfrak{j}_V.$$

Taking quotient on $\ln q$,

$$h(C) - h(J) \leq \frac{\mathfrak{j}_V}{\ln q}.$$

For a dependent set K, let J be the maximal independent set contained in K. Let $k \in K \setminus J$ and $\gamma(k, J)$ be the unique circuit contained in $k \cup J$. By the submodularity of the entropy,

$$h(\gamma(k, J)) + h(J) \geq h(J \cup k) + h(\gamma(k, J) \setminus k),$$

and iterating it,

$$h(K) - h(J) \leq \sum_{k \in K \setminus J} [h(J \cup k) - h(J)] \leq \sum_{k \in K \setminus J} [h(\gamma(k, J) - h(\gamma(k, J) \setminus k)].$$

For every $k \in K \setminus J$, use the bound in for the circuit $\gamma(k, J)$ and independent set $\gamma(k, J) \setminus k$,

$$h(K) - h(J) \leq |Q| \frac{\mathfrak{j}_V}{\ln q}$$

then

$$h(K) - |E|\frac{\mathfrak{j}_V}{\ln q} \leq h(J) \leq |J| = r(K). \tag{10}$$

To get the other bound, from the Lang-Weil inequality

$$h(E) \geq \frac{\ln |V_{\mathbb{F}}|}{\ln q} \geq r(\mathcal{M}) + \frac{\ln(1 - \kappa_V/\sqrt{q})}{\ln q}$$

Now use the bound (10) with $K = E$ and I a base (i.e. a maximal independent subset) and that $h(I) \leq h(J) + h(I \setminus J)$,

$$r(\mathcal{M}) + \frac{\ln(1 - \kappa_V/\sqrt{q})}{\ln q} \leq h(E) \leq h(I) + |E|\frac{\mathfrak{j}_V}{\ln q} \leq h(J) + |I \setminus J| + |E|\frac{\mathfrak{j}_V}{\ln q}$$

since $H(P_{\mathbb{F}}^{I \setminus J}) \leq \ln q^{|I \setminus J|} = |I \setminus J| \ln q$. Using that $r(I) = r(\mathcal{M})$ and that $r(\mathcal{M}) - |I \setminus J| \leq r(K)$, we obtain

$$r(K) + \frac{\ln(1 - \kappa_V/\sqrt{q})}{\ln q} - |E|\frac{\mathfrak{j}_V}{\ln q} \leq h(J) \leq h(K). \tag{11}$$

Combining both bounds, (10) and (11), we get that for every K arbitrary subset of E,

$$\big|h(K) - r(K)\big| \leq \frac{\mathfrak{n}_V}{\ln q}$$

for some constant $\mathfrak{n}_V$ not depending on q. □

Remark C.3. There are several open questions related to this proof in order to define better secret sharing schemes from the polymatroids obtained.

First, the size of the field $\mathbb{F}$ such that the variety $V_{\mathbb{F}}$ is nonempty is unknown, and this size defines later the first field where the sequence of polymatroids is defined. Moreover, the number of points of the variety is estimated by the Lang-Weil bound; however it can be improved with more sophisticated bounds in [31]. This could lead to finding a better approximation of the rank function of the polymatroids $h_{\mathbb{F}}$, and improvements of this bound would lead to better privacy and correctness bounds.

A better knowledge of the polynomials that define the algebraic variety could also give better bounds and improve the convergence of the polymatroids to the original matroid. For example, a good improvement could be a bound for the field $\mathbb{G}_{V_{\overline{\mathbb{G}}}}$ where the coefficients live, or the degree of the polynomials that define the variety.

D Partial Secret Sharing Schemes

For the proof of Theorem 1.8 we construct schemes with relaxed security properties, called partial security. The security notion of these schemes is weaker than the one of perfect and statistical schemes, and only requires that the mutual information between the secret and a subset of parties is higher for subsets in the access structures than for subsets that are not [54]. Next, we introduce a definition of secret sharing schemes in terms of random variables that includes the non-perfect ones; this definition assumes some probability distribution on the secrets.

Definition D.1 (Alternative definition of SSS). *Let E be a finite set of* parties*, let $0 \in E$ be a distinguished party, which is called* dealer*. A* secret sharing scheme *Σ on the set E is a discrete random vector $(S_i)_{i\in E}$ such that $H(S_0) > 0$ and $H(S_0|S_{E\setminus\{0\}}) = 0$.*

In the previous definition, the random variable S_0 corresponds to the *secret value*, while the random variables $(S_i)_{i\in E\setminus\{0\}}$ correspond to the *shares* that are distributed among the parties. For a subset $A \subseteq E \setminus \{0\}$, we define S_A as the jointly distributed random variable that takes the projection of $(S_i)_{i\in E\setminus\{0\}}$ over the coordinates in A.

Let $\Sigma = (S_i)_{i\in E}$ be a secret sharing scheme according to Definition 6.1. We say that Σ is perfectly realized by an access structure Γ if the following correctness and privacy requirements hold:

Correctness: If $A \in \Gamma$, then the random variable S_0 is completely determined by the value of S_A, i.e.,

$$H(S_0|S_A) = 0 \qquad \text{(or equivalently } I(S_0{:}S_A) = H(S_0)\text{)},$$

which implies that the secret value is determined by the shares of the parties in A.

Privacy: For a set $B \notin \Gamma$, the random variables S_0 and S_B are independent, i.e.,

$$H(S_0|S_B) = H(S_B) \qquad \text{(or } I(S_0{:}S_B) = 0\text{)},$$

that is, the shares of the parties in B do not provide any information on the secret in this situation.

This definition is equivalent to the definition of perfect secret sharing schemes given in Definition 2.2 (see [10]).

From this new definition of secret sharing schemes, we can define when an access structure can be partially realized by a secret sharing scheme. The idea of this weaker notion of security is that the amount of information gained about the secret by every authorized set is strictly larger than that of any forbidden one.

Definition D.2 (Partial SSS). *A secret sharing scheme $\Sigma = (S_i)_{i\in E}$ is a* partial scheme *for the access structure Γ (or* partially realizes *Γ) if*

$$C = \min_{B\notin\Gamma} H(S_0|S_B) - \max_{A\in\Gamma} H(S_0|S_A) > 0.$$

This parameter C is called secret capacity*. We say a scheme is* private *if unqualified sets gain no information about the secret, $H(S_0|S_B) = H(S_0)$ for any*

B unqualified set in Γ. We say a scheme is correct *if qualified sets fully recover the secret, $H(S_0|S_A) = 0$ for any A qualified set in Γ.*

The partial information ratio *of the scheme is defined as $\frac{1}{C}\frac{\max_i H(S_i)}{H(S_0)}$. The partial information ratio of an access structure is the infimum of the partial information ratio of all secret sharing schemes that partially realize it.*

E Statistical Schemes from Partial Schemes

In this section, for sake of completeness, we describe the construction from [54] of a statistical secret sharing scheme from a partial secret sharing scheme.

E.1 Wiretap Channel

In this section, we give the definition of a the multi-receiver wiretap channel, which is a generalization of the wiretap channel introduced by Wyner [80]. We give the necessary notations to quantify the reliability and the secrecy of this kind of wiretap channel model, following the work of Jafari and Khazaei [54].

Let $\mathcal{X}$ be the input alphabet and $(\mathcal{Y}_i)_{i\in\mathcal{R}}$, $(\mathcal{Z}_i)_{i\in\mathcal{E}}$ be output alphabets. Let $X, (Y_i)_{i\in\mathcal{R}}, (Z_j)_{j\in\mathcal{E}}$ be $|\mathcal{R}| + |\mathcal{E}| + 1$ jointly distributed random variables with supports $\mathcal{X}$, $(\mathcal{Y}_i)_{i\in\mathcal{R}}$, $(\mathcal{Z}_i)_{i\in\mathcal{E}}$ respectively, where $(\mathcal{Y}_i)_{i\in\mathcal{R}}$ and $(\mathcal{Z}_i)_{i\in\mathcal{E}}$ correspond to the channels outputs of the receivers and eavesdroppers.

A wiretap channel models communication between a sender, a set of legitimate receivers with index set $\mathcal{R}$ and a set of eavesdroppers with index set $\mathcal{E}$. When the sender transmits a message given by the random variable X through the channel, according to a probability distribution p, each receiver $i \in \mathcal{R}$ obtains a message Y_i and each eavesdropper $j \in \mathcal{E}$ gets a message Z_j. The goal of the wiretap channel is to reliably transmit a message to the receivers by using m independent instances of the channel for some $m \in \mathbb{N}$, while keeping it secret from the eavesdroppers. This can be obtained by using two algorithms:

1. A publicly-known probabilistic algorithm for encoding $\mathsf{Enc} : \mathcal{K} \longrightarrow \mathcal{X}^m$
2. Deterministic algorithms for decoding, $\mathsf{Dec}_i : \mathcal{Y}_i^m \longrightarrow \mathcal{K}$, one for every receiver $i \in \mathcal{R}$,

where $\mathcal{K}$ is the set of messages. To transmit a message k in $\mathcal{K}$, the sender encodes it to obtain a tuple $x^m = (x_1, \ldots, x_m) \leftarrow \mathsf{Enc}(k)$. Each symbol x_k is independently distributed through the channel, let $y_i^m = (y_1, \ldots, y_m)$ and $z_j^m = (z_1, \ldots, z_m)$ be the tuples that the receivers and eavesdroppers i and j get respectively. Each receiver uses its own decoder Dec_i to compute a message $\widehat{k_i}$. We denote K as the random variable for the message, in $\mathcal{K}$. Let X^m, Y_i^m, Z_j^m be the encoders output, the receiver and eavesdropper input respectively. And denote $\widehat{K_i}$ the i-th decoder's output. The following definition is from [54].

Definition E.1. *A* rate *$R \geq 0$ is achievable if for every m there exist an encoder* Enc *and decoders* Dec_i*, $i \in \mathcal{R}$, such that:*

1. *K is uniformly distributed on $\mathcal{K} = \{1, \dots, e^{mR}\}$.*
2. ***Reliability.*** *For every receiver $i \in \mathcal{R}$, the average decoding error probability $\mathsf{Pr}(\mathsf{Dec}_i(Y_i^m) \neq K)$ is negligible in m, where the probability is over the uniform distribution on $\mathcal{K}$ and the randomness of the encoding algorithm.*
3. ***Privacy.*** *For every eavesdropper $j \in \mathcal{E}$, the average statistical distance $\mathsf{SD}(p_{Z_j^m K}, p_{Z_j^m} p_K)$ is negligible in m.* [5]

In this case, we say that the set of pairs $(\mathsf{Enc}, \mathsf{Dec}_i)_m$ is a CK rate-R wiretap coding family *for the probability distribution $\Sigma = (X, (Y_i)_{i\in\mathcal{R}}, (Z_j)_{j\in\mathcal{E}})$.*

The secret capacity *of a wiretap channel associated to the distribution $\Sigma = (X, (Y_i)_{i\in\mathcal{R}}, (Z_j)_{j\in\mathcal{E}})$ is the maximum of the rates R of any CK rate-R wiretap coding family for this distribution.*

In general, the secret capacity of the wiretap channel is an open problem. However, it can be proved that if the value

$$C = \min_{i\in\mathcal{R}} I(X : Y_i) - \max_{j\in\mathcal{E}} I(X : Z_j) \tag{12}$$

satisfies $C > 0$, then it is a lower bound on the secret capacity of a wiretap channel associated to Σ (see [81] or [37]).

Note that the conditions of reliability and privacy defined above require that the error probability is negligible on the average; however, if we recall the definition of statistical security for secret sharing, we need to consider stronger requirements:

2'. **Strong reliability.** For every receiver $i \in \mathcal{R}$ and every message $k \in \mathcal{K}$, the decoding error probability $\mathsf{Pr}(\mathsf{Dec}_i(Y_i^m) \neq K | K = k)$ is negligible in m.

3'. **Strong privacy.** For every eavesdropper $j \in \mathcal{E}$ and every message $k \in \mathcal{K}$, the statistical distance $\mathsf{SD}(p_{Z_j^m|K=k}, p_{Z_j^m})$ is negligible in m.

Let $R > 0$ be a fixed achievable rate with respect to the weak requirements. It is known that it is also achievable by strong reliability and privacy requirements [54]. This is done by reducing the message size by a factor of at most $e^{-|\mathcal{R}|+|\mathcal{E}|}$.

Lemma E.2 ([54]). *The rate $R = C - \Theta(\frac{1}{m^{1/4}})$ is achievable with strong reliability and strong privacy requirements with the upper bound $2e^{-\sqrt{m}}$ and $3e^{-\sqrt{m}}$ for reliability and privacy errors respectively.*

[5] The statistical distance between two discrete random variables X and Y, with respective probability mass functions p_X and p_Y is defined as

$$\mathsf{SD}(p_X, p_Y) := \frac{1}{2} \sum_x |Pr(X = x) - Pr(Y = x)|.$$

E.2 Construction of Statistical Schemes from Partial Schemes

Let $\Pi = (S_i)_{i \in E}$ be a partial scheme for the access structure Γ with partial information ratio σ, i.e. recall the Definition D.2. Then, the secret capacity of the scheme is

$$C = H(S_0)\delta = \min_{A \in \Gamma} I(S_0 : S_A) - \max_{B \notin \Gamma} I(S_0 : S_B) > 0.$$

We define a multi-receiver and multi-eavesdropper wiretap channel where each qualified set of the access structure Γ can be viewed as a receiver, and each unqualified set is an eavesdropper. See Appendix E.1 for the definition of a multi-receiver and multi-eavesdropper wiretap channel.

Let

$$\Sigma = (X, (Y_A)_{A \in \Gamma}, (Z_B)_{B \notin \Gamma}) = (S_{P_0}, (S_A)_{A \in \Gamma}, (S_B)_{B \notin \Gamma})$$

and consider the associated wiretap channel. Notice that the secret capacity coincides with the constant C associated to the wiretap channel Σ defined in (12). By Lemma E.2, the rate $R = C - \Theta(\frac{1}{m^{1/4}})$ is achievable.

The wiretap channel that gives this rate is defined as the sequence of random variables $(K_\ell, X^\ell, (S_A)^\ell_{A \in \Gamma}, (S_B)^\ell_{B \notin \Gamma}, (\widehat{K_A})^\ell_{A \in \Gamma})_{\ell \in \mathbb{N}}$ which transmits a message K_ℓ of the space $\{1, \ldots, e^{\ell R}\}$ to the receivers, which are the authorized subsets of Γ. Notice that for convenience, the base of the logarithms of the entropy function is changed to e [54]. The variable X^ℓ is the output of the encoding

$$\mathsf{Enc} : \{1, \ldots, e^{\ell R}\} \to (\mathrm{supp}(S_{P_0}))^\ell.$$

Then, ℓ copies of the random variables $S_1, \ldots, S_n$ are made and every element of the vector X^ℓ is the value of every S_0. For each value of S_0, the rest of the variables $S_1, \ldots, S_n$ are defined, and for each $1 \leq i \leq n$, the vector $S_i^\ell = (S_i^j)_{1 \leq j \leq \ell}$ is defined coordinatewise with the ℓ values of S_i. S_A^ℓ and S_B^ℓ correspond to the receivers and eavesdroppers inputs of the wiretap channel. And finally $\widehat{K_A}^\ell$ is the output of the decoding algorithm of the variable S_A^ℓ defined by the maximum likelihood criterion.

Proposition E.3. *Let Σ be a partial secret sharing scheme with capacity C and information ratio σ. The secret sharing scheme Σ_{stat} with security parameter ℓ shares a secret $k \in \{1, \ldots, e^{\ell R}\}$ and satisfies the following properties:*

$$\mathsf{Pr}(\widehat{K_A}^\ell \neq K_\ell | K_\ell = k) < 2e^{-\sqrt{\ell}} \textit{(Strong reliability)}$$

for every $A \in \Gamma$ and for every $k \in \{1, \ldots, e^{\ell R}\}$, and

$$\mathsf{SD}(p_{S_B^\ell | K_\ell = k}, p_{S_B^\ell}) \leq 3e^{-\sqrt{\ell}} \textit{ (Strong privacy).}$$

for every $B \notin \Gamma$.

The proof of this proposition is deduced from Lemma E.2.

Now, we use the random variables of the wiretap to define a secret sharing scheme that will statistically realize the access structure Γ with security parameter ℓ. Define the sequence of random variables $(T_0^\ell, \dots T_n^\ell)_{\ell \in \mathbb{N}}$ as

$$\begin{aligned} T_0^\ell &= K_\ell, \\ T_i^\ell &= S_i^\ell, 1 \leq i \leq n. \end{aligned}$$

The T_0^ℓ will be the secret random variable and the T_i^ℓ will be the shares. Observe that for every set $A \in \Gamma$, there exists a sequence of random variables $(\widehat{K_A})_{\ell \in \mathbb{N}}$, which is the result of the maximum likelihood algorithm of the variables T_A^ℓ. Proposition 6.2 follows from the properties of the variables given in Proposition E.3.

Let Σ be the statistical scheme with share functions $(T_i^\ell)_{i \in E}$ with security parameter ℓ. Then, the information ratio σ of Σ is

$$\frac{\max_{i \in P} H(T_i^\ell)}{H(T_0^\ell)}.$$

Claim E.4. *Let σ be the partial information ratio of the scheme Π. Then, the information ratio of the scheme Σ tends to σ when ℓ increases.*

Proof. Let i be a qualified party in the access structure. Then,

$$\lim_{\ell \to \infty} \frac{H(T_i^\ell)}{H(T_0^\ell)} = \lim_{\ell \to \infty} \frac{H(S_i^\ell)}{H(K_\ell)} = \lim_{\ell \to \infty} \frac{H(S_i^\ell)}{\ell(C_\Pi - \Theta(\ell^{-1/4})) \log e} = \frac{H(S_i)}{C_\Pi \log e}.$$

Here we have used the relation $\lim_{\ell \to \infty} \frac{H(S_i^\ell)}{\ell} = H(S_i)$, which is known to hold for a wiretap channel [54]. For an unqualified party j, the result is similar. Then,

$$\lim_{\ell \to \infty} \frac{\max_{i \in P} H(T_i^\ell)}{H(T_0^\ell)} = \max_{i \in P} \frac{H(S_i)}{C_\Pi \log e} = \frac{1}{\delta} \frac{\max_{i \in P} H(S_i)}{H(S_{P_0})} = \sigma.$$

$\square$

Summing up, Jafari and Khazaei [54] found a statistical secret sharing scheme Σ for the access structure Γ with information ratio tending to σ when the security parameter increases.

Proof of Proposition 6.2. The proof of this result is based on a construction done in [54], which starts with statistical secret sharing schemes from a set of random variables with positive secret capacity i.e., with $C = \min_{B \notin \Gamma} H(S_0|S_B) - \max_{A \in \Gamma} H(S_0|S_A) > 0$. [54, Theorem 6.1] proves that from this set of random variables with positive secret capacity $S_0, \dots, S_n$, it can be defined a statistical family of secret sharing schemes $(T_0^\ell, \dots, T_n^\ell)_\ell$ with secret space of size $2^{\ell R}$, satisfying Equation (4) and Equation (5). The main tool used to prove this is the wiretap channel; see [54] and Appendix D for more details. Now, consider the secret sharing scheme Π with security parameter ℓ that takes as shares the outcome of the random variables $(T_1^\ell, \ldots, T_n^\ell)_\ell$ and T_0^ℓ as secret, with uniform randomness. It satisfies that for every authorized set A in Γ and every r, s,

$$\mathsf{Pr}(\textsc{Recon}_A(T_A^\ell) = T_0^\ell | T_0^\ell = s) = \mathsf{Pr}(\textsc{Recon}_A(\Pi_\ell(s; r)_A) = s) > 1 - 2e^{\sqrt{-\ell}}.$$

And for every unauthorized set B of Γ and secret s, $\mathsf{SD}(p_{T_B^\ell | T_0^\ell = s}, p_{T_0^\ell}) \leq 3e^{-\sqrt{\ell}}$. Then, by the triangle inequality, for every pair of secret s, s',

$$\mathsf{SD}(T_B^\ell | T_0^\ell = s, T_B^\ell | T_0^\ell = s') = \mathsf{SD}(\Pi_\ell(s; r)_A, \Pi_\ell(s'; r)_A) \leq 6e^{-\sqrt{\ell}}.$$

□

E.3 Statistical Schemes with Information Ratio Tending to 1

Suppose now that we have a sequence of partial schemes $\{\Pi_k\}_k$ with information ratios π_k and they satisfy $\lim_{k\to\infty} \pi_k = 1$. The following lemma is the principal tool to get a statistical scheme with information ratio tending to 1 instead of σ.

Lemma E.5 ([54]). *Let $\{\sigma_i\}_{i\in\mathbb{N}}$ be a sequence of real numbers with $\lim_{i\to\infty} \sigma_i = 1$ and $\{c_i\}_{i\in\mathbb{N}}$ another sequence of real numbers. If a sequence of real numbers $\{\sigma_{i,\ell}\}$ satisfies that $\lim_{\ell\to\infty} \sigma_{i,\ell} = \sigma_i$ and $\{\ell_{i,\ell}\}$ satisfies that $\ell_{i,\ell} \leq c_i \ell$, then there exists $\mu : \mathbb{N} \to \mathbb{N}$ such that*

$$\lim_{\ell\to\infty} \sigma_{\mu(\ell),\ell} = 1, \qquad \textit{and} \qquad \ell_{\mu(\ell),\ell} < \ell^2.$$

Proof. For every i, since $\lim_{\ell\to\infty} \sigma_{i,\ell} = \sigma_i$, there exists some M_i for which $|\sigma_{i,\ell} - \sigma_i| \leq 1/i$ for every $\ell \geq M_i$. We want $\mu(\ell)$ to satisfy $\ell > M_{\mu(\ell)}$ and also, we want $\ell > c_{\mu(\ell)}$, so let

$$d_i = \max\{M_1, \ldots, M_i, c_1, \ldots, c_i\} + i.$$

Observe that d_i is an increasing sequence of non-negative real numbers and it goes to infinity as $i \to \infty$. Then, define $\mu(\ell)$ as follows:

$$\mu(\ell) = \begin{cases} 1 \text{ if } \ell < d_1, \\ i \text{ if } d_i < \ell \leq d_{i+1} \end{cases}$$

Note that $\ell > d_{\mu(\ell)} \geq M_{\mu(\ell)}$, so $|\sigma_{\mu(\ell),\ell} - \sigma_{\mu(\ell)}| \leq 1/i$. Since $\mu(\ell)$ is also a monotone increasing unbounded sequence, $\lim_{\ell\to\infty} \sigma_{\mu(\ell)} = 1$. Then, we have that

$$\lim_{\ell\to\infty} \sigma_{\mu(\ell),\ell} = 1.$$

Also note that since $\ell > d_{\mu(\ell)} \geq c_{\mu(\ell)}$, we have that $\ell_{\mu(\ell),\ell} \leq c_{\mu(\ell)}\ell < \ell^2$. □

Let Σ_k be the statistical scheme constructed from the partial secret sharing scheme Π_k as in Proposition 6.8. Let σ_k be the information ratio of every scheme; by Claim E.4 σ_k tends to π_k, the information ratio of the partial scheme. Let C_{Π_k} be the secret capacity of the partial scheme Π_k, we also want that the secret length of the final family grows polynomially in ℓ, so let $\ell_{k,\ell}$ be the secret length of the scheme Σ_k, note that it satisfies that

$$\ell_{k,\ell} \leq c_{\Pi_k} \ell.$$

For every ℓ, we want a value $\mu(\ell)$ for which the secret sharing scheme $\Sigma_{\mu(\ell)}$ will be selected. This value has to satisfy that $\lim_{\ell \to \infty} \sigma_{\mu(\ell)} = 1$ and $\ell_{\mu(\ell),\ell} = O\left(\ell^c\right)$ for some $c > 0$. With this value, for each ℓ we will consider the statistical sharing scheme $\Sigma_{\mu(\ell)}$ with security parameter ℓ.

Note that the statistical correctness and privacy still hold for the scheme $\Sigma_{\mu(\ell)}$ since the errors of the scheme Σ_k do not depend on k and then, the errors of the final scheme $\Sigma_{\mu(\ell)}$ are still negligible in ℓ. Finally, we obtained a statistical secret sharing scheme, for the access structure Γ with information ratio tending to 1 with the security parameter and with quadratic secret length growth. This completes the proof of the Theorem 1.9.